Automated Assessment of Residual Plots with Computer Vision Models

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Abstract

~~Plotting the residuals is a recommended procedure to diagnose deviations from linear model assumptions, such as non-linearity, heteroskedasticity, and non-normality. The presence of structure in residual plots can be tested using the lineup protocol to do visual inference. There are a variety of conventional residual tests, but the lineup protocol , used as~~ Residual plots are routinely used to diagnose linear regression models, but their interpretation relies on human judgement and can be misleading. The lineup protocol provides a statistical test ~~, performs better for diagnostic purposes because it is less sensitive and applies more broadly to different types of departures. However, the lineup protocol relies on human judgment which limits its scalability. This work presents a solution by providing a computer vision model to automate the assessment of residual plots. It is trained to predict~~ of visual discoveries, but requires a panel of human viewers and is difficult to scale. We propose an automated method combining computer vision with statistically valid hypothesis testing. We define a distance measure ~~that quantifies the disparity between~~ quantifying the departure of the residual distribution ~~of a fitted classical normal linear regression model and the reference distribution, based on Kullback-Leibler divergence. Evaluation using data from a prior human-subject experiment shows the model exhibits lower sensitivity than conventional tests but higher sensitivity than human visual tests. Several examples from classical papers and contemporary data demonstrate the method 's practical utility in automating residual diagnostics~~under a potentially misspecified model from that expected under correct specification. A convolutional neural network predicts this distance directly from residual plot images, learning visual features associated with model departures. Statistical significance is assessed by comparing the predicted distance of the observed plot with those of null plots generated through rotations of the observed residuals. We demonstrate accurate distance prediction on held-out data and show that the resulting test exhibits rejection rates between those of conventional residual-based tests and human viewers, depending on the severity of departures. Three examples illustrate how the method discounts visually salient features that also arise under the null while identifying visual patterns that conventional tests may overlook. The proposed method provides a scalable means of screening residual plots, helping readers determine where closer visual inspection is most needed.

Keywords: statistical graphics, data visualization, visual inference, computer vision, machine learning, hypothesis testing

1. Introduction

Plotting residuals is ~~commonly regarded as a standard practice in linear regression diagnostics (Belsley et al., 1980; Cook . This visual assessment plays a crucial role in identifying whether key model assumptions, such as linearity,~~

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homoscedasticity, and normality, are reasonable. It also helps in understanding the goodness of fit and various unexpected characteristics of the model.

Generating standard practice after fitting a linear regression (Belsley et al., 1980; Cook and Weisberg, 1982) . Its enduring value lies in revealing *where* a model fails rather than merely *whether* it does. Such failures may manifest as curvature in the mean, a fan-shaped spread, or a few influential observations. Producing a residual plot in most statistical software is often as straightforward as executing a takes a single line of code or clicking a button. However, accurately interpreting a residual plot can be challenging, as some features are a result of predictor characteristics and natural variation, while others indicate violations of model assumptions (Li et al., 2024). For example, in Figure 1 , the residual plot displays a triangular in any statistical environment. Interpreting it, however, is left to the analyst, and that is where the difficulty begins.

Residual plots can be ambiguous, and seemingly compelling patterns do not always indicate a model violation. Figure 1 illustrates a residual plot in which the spread narrows as fitted values increase, forming a left-pointing shape. The distinct difference in the spread of the residuals across the fitted values might lead analysts to suspect heteroskedasticity. In this case, however, the fitted regression triangle. This pattern naturally suggests heteroskedasticity, and the Breusch–Pagan test appears to confirm that interpretation, rejecting the null hypothesis at the 5% level ($p = 0.046$). However, the model is correctly specified, and the triangular shape is actually a result of the . The apparent pattern arises from a skewed distribution of the predictors , rather than indicating a flaw in the model . rather than a violation of the model assumptions (Li et al., 2024). In this case, visual inspection and formal testing agree, yet both lead to a false conclusion. More generally, interpreting residual plots is challenging not only because visual patterns can be misleading, but also because their interpretation depends on human judgement, which can vary across analysts.

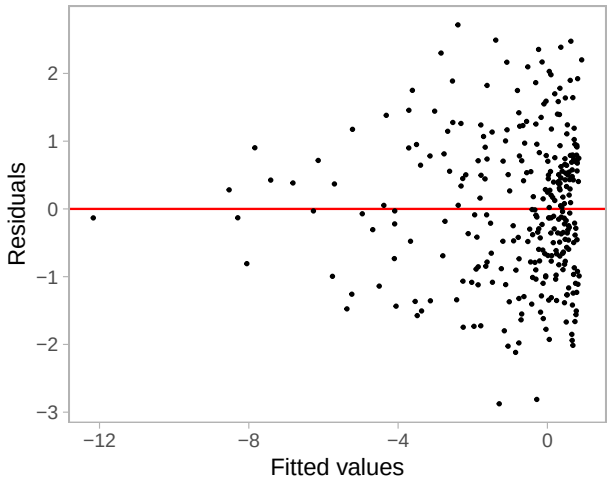
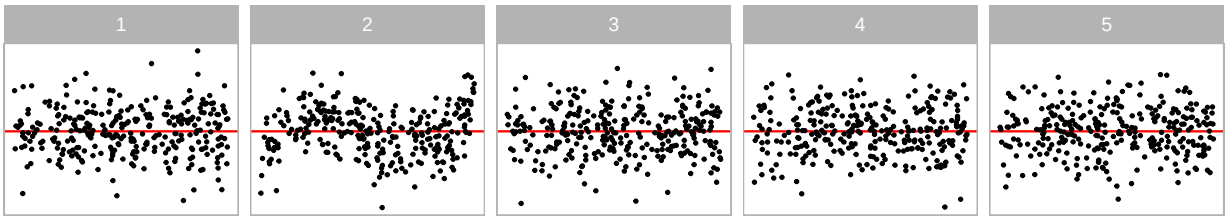


Figure 1: An example residual vs residuals versus fitted values plot (red line indicates 0 corresponds to the x-intercept, i.e. $y = 0$). The vertical spread of the data points varies with the fitted values. This often indicates the existence of heteroskedasticity, however, here the result is due to skewed distribution of the predictors rather than heteroskedasticity. The Breusch-Pagan test rejects this residual plot at 5% significance level (p-value = 0.046).

The concept of visual inference, as proposed by Buja et al. (2009) , provides an inferential framework for assessing whether residual plots contain visual patterns inconsistent with the model assumptions. In this framework, a **true residual plot** refers to the residual plot obtained from the observed data under the fitted regression model. In contrast, Buja et al. (2009) addressed both challenges by introducing the lineup protocol, which turns plot interpretation into a statistical procedure, a form of visual inference. The observed residual plot is hidden among m **null plots** are synthetic residual plots generated under the null hypothesis generated under H_0 , where the linear regression model is correctly specified. These null plots are constructed by generating residuals from the residual rotation distribution (Langsrud, 2005), which

preserves the structure implied by H_0 (see Section 4 for details).

The goal is to assess whether the true residual plotis visually distinguishable from the null plots. This is typically done using the lineup protocol (Figure 2), where the true residual plot is embedded among multiple null plots . Successful identification of by rotating the observed residuals (Langsrud, 2005). A panel of viewers is asked to identify the most different plot. Under H_0, all $m+1$ plots are exchangeable, so each viewer has a probability of $1/(m+1)$ of selecting the observed plot. The proportion of viewers who identify it provides the basis for a p-value. Figure 2 illustrates a lineup with one observed plot and four null plots. The protocol has been widely used in applied work (Loy and Hofmann, 2013, 2014, 2015; Widen et al., 2016; Fieberg et al., 2024 . However, each lineup requires a panel of human judges, limiting its scalability for the true residual plotprovides evidence against H_0. large numbers of models now being fitted and diagnosed. Automating plot interpretation could overcome this limitation, but requires methods capable of recognising a wide range of diagnostic patterns.



An example lineup embedding the true residual plot among four null plots. In practice, lineups typically include 19 null plots, but a reduced set is shown here for presentation purposes. The null plots are generated via residual rotation to ensure consistency with H_0. Observers who have not previously seen the lineup are asked to identify the plot that appears most different. Under H_0 that the regression model is correctly specified, the true residual plot should be indistinguishable from the null plots, yielding a selection probability of 0.2. A small p-value arises when a substantial proportion of observers select the true residual plot (shown at position 2, exhibiting non-linearity).

Figure 2: An example lineup embedding the observed residual plot among four null plots. In practice, lineups typically include 19 null plots, but a reduced set is shown here for presentation purposes. The null plots are generated via residual rotation to ensure consistency with H_0. Observers who have not previously seen the lineup are asked to identify the plot that appears most different. Under H_0 that the regression model is correctly specified, the observed residual plot should be indistinguishable from the null plots, yielding a selection probability of 0.2. A small p-value arises when a substantial proportion of observers select the observed residual plot (shown at position 2, exhibiting non-linearity).

The practice of delivering a residual plot as a lineup is generally regarded as a valuable approach. Beyond its application in residual diagnostics, the lineup protocol has been integrated into the analysis of diverse subjects. For instance, Loy and Hofmann (2013; 2014; 2015) illustrated its applicability in diagnosing hierarchical linear models. Additionally, Widen et al. (2016) and Fieberg et al. (2024) demonstrated its utility in geographical and ecology research respectively, while Krishnan and Hofmann (2021) explored its effectiveness in forensic examinations.

A practical limitation of the lineup protocol lies in its reliance on human judgments (see Li et al., 2024) . Unlike conventional statistical tests that can be performed using a statistical software, the lineup protocol requires human evaluation of images. This characteristic makes it less suitable for large-scale applications, given the associated high labor costs and time requirements. There is a substantial need to develop an approach to substitute these human judgement, for example through the automated interpretation of data plots using computer vision.

Efforts Early efforts to automate plot interpretation date back to early work like include Tukey's "Scagnostics " (Tukey and Tukey, 1985), which is Scagnostics (Tukey and Tukey, 1985), a set of numerical statistics that summarize features of scatter plots. Wilkinson et al. (2005) expanded on this work, introducing scagnostics based on computable measures applied to summarise the features of a scatter plot. Wilkinson et al. (2005) extended this idea with measures computed from planar proximity graphs. These measures, including , but not limited to, , including "Outlying", "Skinny", "Stringy", "Straight", "Mono-

tonic", "Skewed", "Clumpy" ~~,~~ and "Striated", ~~aimed to characterize~~ which characterise outliers, shape, density, trend ~~, coherence and other characteristics of the data~~and coherence. However, ~~such measures may not capture the full range of visual patterns necessary for residual diagnostics (Buja et al., 2009). A more promising direction lies in enabling computers to learn the relevant visual features directly mirroring how humans intuitively interpret plots.~~ these measures were developed to characterise general scatterplot features rather than assess regression model adequacy. Moreover, a fixed dictionary of features may fail to capture the range of patterns that residual plots can exhibit (Buja et al., 2009).

~~Modern computer vision models rely on deep neural networks with convolutional layers (Fukushima and Miyake, 1982) . These layers use small, sliding windows to scan the image, performing a dot product to extract local features and patterns . Numerous studies have demonstrated the efficacy of convolutional layers in addressing various vision tasks, including image recognition (Rawat and Wang, 2017). Despite the widespread use of computer vision models in fields like computer-aided diagnosis (Lee and Chen, 2015), pedestrian detection (Brunetti et al., 2018), and facial recognition (Emami and Suciu, 2012), their application in reading data plots remains limited. While some studies have explored the use of computer vision models for tasks such as reading recurrence plots for~~ An alternative is to learn relevant features directly from plot images using convolutional layers that detect local patterns (Fukushima and Miyake, 1982). Computer vision methods have been used to interpret recurrence plots in time series regression (Ojeda et al., 2020), time series classification ~~(Chu et al., 2019; Hailesilassie, 2019; Hatami et al., 2018; Zhang et al., 2020), anomaly detection (Chen et al., 2020) , and pairwise causality analysis (Singh et al., 2017), the application of reading residual plots with computer vision models is a new field of study.~~ (Chen et al., 2020) and anomaly detection (Singh et al., 2017), but their application to residual plot assessment remains unexplored. However, recognising visual patterns is not sufficient for statistical diagnosis. An effective diagnostic method must quantify the severity of departures from model assumptions and assess whether those departures are statistically significant.

~~In this paper, we develop computer vision models and integrate them into the residual plot diagnostics workflow, addressing the need for automated visual inference. The model is a regression-based computer vision system that takes a residual plot as input and outputs a non-negative estimated distance $\hat{D}$. Training and test data are generated from a synthetic data model, simulating residual plots under various violation scenarios, with targets defined as customized distances D computed from the underlying data generating process, quantifying the extent of model violations. The model is evaluated on held-out test plots for its ability to estimate $\hat{D}$. In addition, we develop a statistical testing procedure based on $\hat{D}$ for detecting model misspecification, and compare its performance with classical residual diagnostic tests as well as results from a human-subject visual inference experiment using the lineup protocol.~~ To quantify severity, we define a distance measure based on the Kullback–Leibler (KL) divergence of the residual distribution under a potentially misspecified model from that under a correctly specified model. We train a convolutional neural network to predict this distance, which represents the theoretical severity, directly from residual plot images, allowing the network to learn visual features relevant to the magnitude of the departure.

~~The paper is structured as follows. Section ?? discusses different specifications of the computer vision models. Section ?? introduces the distance measure used to quantify model violations, while Section ?? describes how this measure is estimated using computer vision models, including the model architecture, data generation, and training procedure. Section 4 describes statistical tests based on the estimated distance, and Section ?? presents bootstrap-based methods for assessing the variability of estimated distance and the stability of model misspecification conclusions. Results are reported in Section ??, followed by illustrative applications on example datasets in Section 6. Finally, we conclude with a discussion of the findings and outline directions for future research.~~

2. ~~Model Specifications~~

~~There are various specifications of the computer vision model that can be used to assess residual plots~~To assess statistical significance, we compare the predicted distance of the observed residual plot with those of null plots generated by rotating the observed residuals. These rotations preserve the fitted values and residual magnitudes, making the observed plot and its rotations exchangeable under H_0. Comparing their

predicted distances therefore yields a test with exact Type I error control. Crucially, the validity of the test does not depend on the accuracy of the learned distance measure. Prediction accuracy affects power, not validity. ~~We discuss these specifications below focusing on the input and the output formats.~~

1.1. ~~*Input Formats*~~

~~Deep learning models are sensitive to input design, and several alternatives are available for encoding residual plots. A simple approach involves feeding vectors of residuals and fitted values into the model. While this format contains all relevant information, the input length varies with sample size, which conflicts with the fixed-shape requirements of modern vision models such as those implemented in TensorFlow (Abadi et al., 2016). Padding can be used to enforce uniform length but risks truncation if inputs exceed the fixed size. Alternatively, fixed-size sampling of residual–fitted value pairs can ensure consistency, though at the cost of some information loss~~The goal of this study is to develop an automated method for assessing residual plots by combining computer vision with statistically valid hypothesis testing. The method learns visual features associated with the severity of departures from model assumptions and uses a rotation-based test to determine whether these departures are statistically significant. Because the aim is to capture the visual patterns that lead human viewers to identify the observed plot among its nulls, we evaluate the method by comparing its decisions with human judgements on lineups, assessing how closely they align.

~~Another strategy is to convert residual plots into images. This enables the use of standard convolutional architectures in computer vision such as VGG16 (Simonyan and Zisserman, 2014). Image-based inputs offer a richer representation by preserving the joint distribution of residuals and fitted values in a visually interpretable format, helping models more effectively capture spatial structure and local correlations than vector-based formats. To ensure model generalization, it is essential to use residual plots in a consistent style, i.e., same aesthetics, layout, scaling and color schemes.~~

~~Using multiple residual plots as input, such as pairs, triplets, or full lineups is also a possible option. For instance, triplet networks (Chopra et al., 2005) can assess visual similarity if we assume null plots are more similar to each other than to the true residual plot , which may help distinguish null plots from true residual plots. However, these architectures introduce complexity in training due to weight sharing and specialized loss functions. We experimented with this setting in a pilot study, and found multiple residual plots led to high input resolution, high computational cost and suboptimal model performance. Balancing accuracy, interpretability, and implementation cost, we adopt a single residual plot in image format as the model input.~~

1.1. ~~*Output Formats*~~

~~Given the input is a single fixed-resolution image, the model output can be defined for various tasks, including binary classification, multiclass classification, and numeric regression.~~

~~In the binary case, the outcome may indicate whether the input image is consistent with a null plot, as determined either by (1) the data-generating process or (~~The remainder of the paper is organised as follows. Section 2 ~~) the outcome of a visual test based on human judgment. The first approach models the probability that a residual plot is not a null plot and is a viable option. In contrast, the second relies on experimental data that are often scarce or inconsistent across studies.~~

~~Alternatively, a multiclass outcome may classify images into categories such as "contains outliers" vs. "does not contain outliers" or "is non-linear" vs. "not non-linear, " among other diagnostic features. While this approach is appealing in theory, it essentially reverts to the traditional testing framework, where tests focus on detecting specific types of model violations in isolation.~~

~~A third option is to produce a meaningful and interpretable numerical measure that quantifies the strength of suspicious visual patterns reflecting the extent of model violations, or the difficulty index for identifying whether a residual plot has no issues. However, such measures are often used informally in daily communications but are rarely formalized. For supervised learning, they must be clearly defined and reliably estimable.~~

~~In this study, we formulate the output as a continuous measure of deviation from model assumptions. Compared to probability outputs based on binary labels (true residual plots labeled as 1 and null plots as~~

0), this formulation explicitly encodes the severity of violations and avoids forcing the model to infer such structure solely from the data distribution, which can be inefficient and discards useful prior knowledgedefines the distance between the residual distribution of a fitted model and the reference distribution it is compared with. Section 3 describes how the distance is predicted from a residual plot, how each training label is computed, and the network and training procedure. Section 4 presents the hypothesis test and establishes its Type I error control. Section 5 reports how accurately the model predicts the distance and compares its decisions with those of viewers and of conventional tests. Section 6 applies the method to three datasets and explores which parts of a plot drive its predictions, Section 7 discusses what the measure cannot see, and Section 8 concludes.

2. Distance from a Theoretically "Good" Residual PlotMeasuring residual plot differences through distributional comparisons

A numerical measure of "difference" or "distance " between plots can take the form of a basic statistical operation on pixels, such as the sum of squared differences distance between two plots can be defined by summing the squared differences between corresponding pixels. However, direct pixel-to-pixel comparisons are not well pixel-based comparisons are poorly suited to residual plots, where the primary interest lies in structural patterns . Alternatively, the distance can be based on established image similarity metricsrather than the precise alignment of individual points. Established image metrics, such as the Structural Similarity Index Measure (Wang et al., 2004), which compares images using contrast, luminance, and structure (related to the average, standard deviation, and correlation of pixel values within a window, respectively). These metrics structural similarity index (Wang et al., 2004), compare luminance, contrast and local structure, but are designed for general image comparison tasks that differ substantially from evaluating rather than data plots, where only essential plot elements are relevant (Chowdhury et al., 2018). Another direction is to define distance using key scatter plot certain visual elements carry statistical information (Chowdhury et al., 2018) . Alternatively, a plot can be summarised using numerical characteristics, such as those captured by scagnostics (see Section 1), although the functional form must be carefully designed to reflect model violationsmeaningfullyScagnostics (Tukey and Tukey, 1985; Wilkinson et al., 2005), and the distance between these summaries can be measured. However, such summaries and their associated distance measures must be chosen to capture patterns indicative of model violations.

In this section, we introduce a distance measure between a true residual plot and a theoretically "good" residual plot. This measure quantifies the divergence between the Rather than comparing rendered images or a fixed set of plot characteristics, we take a distribution-based approach. A residual plot represents a sample from the residual distribution of a given fitted regression modeland that of a correctly specified model. Its construction assumes knowledge of the data generating processes for both predictors and response variables. Since these processes are often unknown in practice, Section ?? discusses how the distance can be estimated using a computer vision model, while Section 4 describes a statistical test based on the estimated distancethe fitted model. We quantify how far this distribution deviates from a reference distribution, which describes the residual behaviour expected under a correctly specified model.

2.1. Residual Distributiondistributions

Consider the classical normal linear regression model, $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{e}$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{e}$, where $\boldsymbol{y}$ is the $n \times 1$$n \times 1$ response vector, $\boldsymbol{X}$ is the design matrix$n \times k$ design matrix of rank k, $\boldsymbol{\beta}$ is a the $k \times 1$ vector of regression coefficients, and $\boldsymbol{e}$ is the vector of random errors. The residual vector is $\hat{\boldsymbol{e}} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{R}\boldsymbol{e}$, where $\boldsymbol{R} = \boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$ is the residuals are $\hat{\boldsymbol{e}} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{R}\boldsymbol{e}$, where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$ is the hat matrix, $\boldsymbol{R} = \boldsymbol{I}_n - \boldsymbol{H}$ is the residual operator, and $\boldsymbol{I}_n$ is the $n \times n$ $n \times n$ identity matrix. Since $\boldsymbol{R}$ is symmetric and idempotent, the residuals lie in the $(n-k)$-dimensional subspace orthogonal to the column space of $\boldsymbol{X}$.

Under the classical assumptions $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}_n, \sigma^2 \boldsymbol{I}_n)$assumption $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}_n, \sigma^2 \boldsymbol{I}_n)$, the residuals are also Gaussian, since they are a linear transformation of $\boldsymbol{e}$. We refer to this induced distribution as the reference distribution Q, which characterizes the residual behaviour under correct model specification and depends

only on $\boldsymbol{X}$. follow a Gaussian distribution with mean zero and covariance $\sigma^2\boldsymbol{R}$. We call this the **reference residual distribution**.

If $\boldsymbol{X}$ has full column rank k, then $\operatorname{rank}(\boldsymbol{R}) = n - k$, and $\operatorname{cov}(\hat{\boldsymbol{e}}) = \sigma^2\boldsymbol{R}$ is singular. Hence,

$$Q = \mathcal{N}\left(\mathbf{0}_n, \sigma^2\boldsymbol{R}\right).$$

Since $\boldsymbol{R}$ has rank $n-k$, Q is a degenerate Gaussian supported on an $(n-k)$-dimensional subspace. To obtain a non-degenerate reference distribution required for the distance computation in Section **??**, we replace $\sigma^2\boldsymbol{R}$ with its diagonal approximation, $\operatorname{diag}(\sigma^2\boldsymbol{R})$, yielding $Q \approx \mathcal{N}\left(\mathbf{0}_n, \operatorname{diag}(\sigma^2\boldsymbol{R})\right)$degenerate in $\mathbb{R}^n$. Its i-th marginal has variance $\sigma^2 R_{ii} = \sigma^2(1 - h_{ii})$, where h_{ii} is the leverage of observation i. Thus, residuals at high-leverage points are less variable than those at other points.

The reference distribution Q corresponds to the correct model specification. Under misspecificationWhen the model is misspecified, the residuals follow a different distribution the **actual residual distribution** P. For example, omitted variables correlated with $\boldsymbol{X}$ alter the residual structure, causing P to deviate from Q. Similarly, non-Gaussian error distributions (e.g. multivariate lognormal) induce skewness and heavier tailsUnlike Q, this distribution need not be Gaussian, centred at zero, or have covariance $\sigma^2\boldsymbol{R}$. Like Q, it is supported on the column space of $\boldsymbol{R}$ and is therefore degenerate in $\mathbb{R}^n$.

2.2.

We define the distance of the true residual distribution Neither Q nor P from the reference residual distribution admits a density with respect to n-dimensional Lebesgue measure, so they cannot be compared directly using a density-based divergence. We therefore map each joint distribution to the product of its coordinatewise marginal distributions. For a distribution F on $\mathbb{R}^n$, let F_i denote its i-th marginal distribution and define

$$\mathcal{M}(F) = \bigotimes_{i=1}^{n} F_i,$$

where $\otimes$ denotes the product of probability distributions. The operator preserves every coordinatewise marginal while making the coordinates independent. Applying it to Q based on the Kullback–Leibler divergence (Kullback and Leibler, 1951):

$$D = \log\left(1 + D_{KL}(P\|Q)\right),$$

gives

$$\mathcal{M}(Q) = \mathcal{N}\left(\mathbf{0}_n, \operatorname{diag}(\sigma^2\boldsymbol{R})\right),$$

where $D_{KL}(P\|Q)$ is defined as

$$D_{KL}(P\|Q) = \int_{\mathbb{R}^n} \log\frac{p(\hat{\boldsymbol{e}})}{q(\hat{\boldsymbol{e}})}\, p(\hat{\boldsymbol{e}})\, d\hat{\boldsymbol{e}},$$

and $p(\cdot)$ and $q(\cdot)$ are the probability density functions for where $\operatorname{diag}(\boldsymbol{A})$ denotes the diagonal matrix formed from the diagonal entries of $\boldsymbol{A}$. Unlike Q, $\mathcal{M}(Q)$ is nondegenerate and admits a density with respect to n-dimensional Lebesgue measure, provided every marginal variance is positive. We similarly obtain $\mathcal{M}(P)$ by applying the same operator to $P$and Q, respectively. .

This measure was introduced in Li et al. (2024) to quantify the effect size of non-linearity and heteroskedasticity in residual plots. In the presence of model misspecification, if the error structure deviates from $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}_n, \sigma^2\boldsymbol{I}_n)$, for instance with covariance $\boldsymbol{V}$ and omitted higher-order terms $\boldsymbol{Z}\boldsymbol{\beta}_z$ correlated with $\boldsymbol{X}$, the induced residual distribution can be expressed as $P = \mathcal{N}(\boldsymbol{R}\boldsymbol{Z}\boldsymbol{\beta}_z, \operatorname{diag}(\boldsymbol{R}\boldsymbol{V}\boldsymbol{R}))$, where the diagonal approximation ensures non-degenerate covariance matrix. The transformation $D = \log(1 + D_{KL})$ compresses large values of D_{KL}, improves numerical stability during estimation (Section **??**), and ensures that D is well-defined when $D_{KL} = 0$We compare $\mathcal{M}(P)$ with $\mathcal{M}(Q)$ rather than comparing P directly with Q. Since the operator

$\mathcal{M}$ retains only the coordinatewise marginal distributions, $P = Q$ implies $\mathcal{M}(P) = \mathcal{M}(Q)$, but the converse does not necessarily hold. Consequently, the comparison captures differences in the location, spread, or shape of individual residual distributions, but not differences in their dependence structure, as discussed in Section 7.

Since P and Q are taken as non-degenerate Gaussian

2.2. Distance of $\mathcal{M}(P)$ from $\mathcal{M}(Q)$

We compare the marginal products through their Kullback–Leibler divergence (Kullback and Leibler, 1951) and define the distance measure

$$D = \log\left(1 + D_{KL}\left(\mathcal{M}(P) \,\|\, \mathcal{M}(Q)\right)\right). \tag{1}$$

Because both arguments are products of n marginal distributions, the KL divergence admits the closed form divergence factorises exactly,

$$D_{KL}\left(\mathcal{M}(P) \,\|\, \mathcal{M}(Q)\right) = \frac{1}{2}\int_{\mathbb{R}^n} \log \frac{|\boldsymbol{W}|}{|\mathrm{diag}(\boldsymbol{R}\sigma^2)|} - n + \mathrm{tr}\,\frac{\prod_{i=1}^n p_i(\hat{e}_i)}{\prod_{i=1}^n q_i(\hat{e}_i)} \prod_{i=1}^n p_i(\hat{e}_i)\, d\hat{\boldsymbol{e}} = \sum_{i=1}^n D_{KL}(\sigma^2 p_i \,\|\, q_i)) + \boldsymbol{z}^{\top}\boldsymbol{z}, \tag{2}$$

where $\boldsymbol{\mu}_z = \boldsymbol{R}\boldsymbol{Z}\boldsymbol{\beta}_z$, and $\boldsymbol{W} = \mathrm{diag}(\boldsymbol{R}\boldsymbol{V}\boldsymbol{R})$. The assumed error variance σ^2 is set to be $\mathrm{tr}(\boldsymbol{V})/n$, which is the expectation of the estimated variance. p_i and q_i are the marginal densities of $\hat{e}_i$ under P and Q. The n-dimensional integral therefore never has to be evaluated.

2.3.

When the error vector $\boldsymbol{e}$ is non-Gaussian, The logarithm is strictly increasing, so D ranks departures in the same order as D_{KL} while compressing large divergence values, which can reduce the influence of extreme target values during neural network training (Section 3). When the induced residual distribution P is generally non-Gaussian and typically unavailable in closed form. Consequently, the Gaussian closed-form expression in Equation ?? is no longer applicable, and the KL divergence must instead be evaluated from its definition in Equation ??. This requires high-dimensional integration over the residual space, rendering direct evaluation impractical for large n. two distributions coincide, $D_{KL} = 0$ and hence $D = 0$.

To obtain a tractable approximation, we assume that the residual components $\{\hat{e}_i\}_{i=1}^n$ are mutually independentAn explicit expression for D requires the marginal densities p_i and q_i. Consider the Gaussian misspecification in which the model omits a term $\boldsymbol{Z}\boldsymbol{\beta}_z$ correlated with $\boldsymbol{X}$, where $\boldsymbol{Z}$ is $n \times q$ and $\boldsymbol{\beta}_z$ is $q \times 1$, and in which the errors have covariance $\boldsymbol{V}$, an $n \times n$ positive definite matrix. If the errors remain Gaussian, both the omitted term and the covariance pass through the residual operator, so that $p(\hat{\boldsymbol{e}}) \approx \prod_{i=1}^n p_i(\hat{e}_i)$ and $q(\hat{\boldsymbol{e}}) \approx \prod_{i=1}^n q_i(\hat{e}_i)$. This assumption holds for $Q \approx \mathcal{N}\left(\boldsymbol{0}_n, \mathrm{diag}(\sigma^2\boldsymbol{R})\right)$, which has independent components, but generally does not hold for P. Nevertheless, it removes the need for high-dimensional integration and reduces the computation to univariate densities , substantially lowering computational cost.

Under this factorisation , the KL divergence decomposes additively:

$$D_{KL}(P\|Q) \approx \sum_{i=1}^n \int_{\mathbb{R}} p_i(\hat{e}_i) \log \frac{p_i(\hat{e}_i)}{q_i(\hat{e}_i)} d\hat{e}_i = \sum_{i=1}^n \mathbb{E}_{p_i}\left[\log \frac{p_i(\hat{e}_i)}{q_i(\hat{e}_i)}\right],$$

$P = \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{R}\boldsymbol{V}\boldsymbol{R})$ with $\boldsymbol{\mu}_z = \boldsymbol{R}\boldsymbol{Z}\boldsymbol{\beta}_z$ and

$$\mathcal{M}(P) = \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{W}), \qquad \boldsymbol{W} = \mathrm{diag}(\boldsymbol{R}\boldsymbol{V}\boldsymbol{R}).$$

With $\boldsymbol{S} = \mathrm{diag}(\sigma^2\boldsymbol{R})$ for $\mathcal{M}(Q)$, where p_i and q_i denote the marginal densities of $\hat{e}_i$ under $\sigma^2 = \mathrm{tr}(\boldsymbol{R}\boldsymbol{V})/(n-k)$ is the residual variance the fitted model would estimate, both arguments are Gaussian with diagonal covariance and

$$D_{KL}\left(\mathcal{M}(P) \,\|\, \mathcal{M}(Q)\right) = \frac{1}{2}\left(\log|\boldsymbol{S}| - \log|\boldsymbol{W}| - n + \mathrm{tr}\left(\boldsymbol{S}^{-1}\boldsymbol{W}\right) + \boldsymbol{\mu}_z^{\top}\boldsymbol{S}^{-1}\boldsymbol{\mu}_z\right). \tag{3}$$

Within this Gaussian misspecification, Equation 3 is exact for the two distributions being compared. Its only approximation is the one carried by $\mathcal{M}$, which replaces the joint dependence between residuals with independence.

Equation 3 is available only when P ~~and Q, respectively.~~

~~We approximate the expectation via~~ is Gaussian and its mean and covariance are known. Outside that family the marginal densities of P do not in general admit a closed form, so Equation 3 cannot be evaluated. The factorisation in Equation 2 still holds exactly, because it follows from the definition of $\mathcal{M}$, and each summand is the divergence of p_i from q_i, where only p_i lacks a closed form. We therefore evaluate the divergence using Monte Carlo simulation. Specifically, we generate ~~independent samples $\{\boldsymbol{e}^{(j)}\}_{j=1}^{m}$ from the error distribution~~ m independent responses $\{\boldsymbol{y}^{(j)}\}_{j=1}^{m}$ from the data generating process and obtain residual vectors ~~$\boldsymbol{B}_j = \boldsymbol{R}\boldsymbol{e}^{(j)}$. The~~ $\boldsymbol{B}_j = \boldsymbol{R}\boldsymbol{y}^{(j)}$. For each coordinate i, we approximate the marginal density p_i ~~is then estimated from~~ from the simulated values $\{B_{ij}\}_{j=1}^{m}$ using a Gaussian kernel density estimator with ~~bandwidth selected by~~ Silverman's rule of thumb (Silverman, 2018), where B_{ij} denotes the i-th component of $\boldsymbol{B}_j$. ~~This yields~~

$$\sum_{i=1}^{n} \mathbb{E}_{p_i}\left[\log \frac{p_i(\hat{e}_i)}{q_i(\hat{e}_i)}\right] \approx \sum_{i=1}^{n} \frac{1}{m} \sum_{j=1}^{m} \log \frac{\hat{p}_i(B_{ij})}{q_i(B_{ij})},$$

We then compute the Monte Carlo approximation to the KL divergence as

$$D_{KL} \approx \sum_{i=1}^{n} \frac{1}{m} \sum_{j=1}^{m} \log \frac{\hat{p}_i(B_{ij})}{q_i(B_{ij})}, \tag{4}$$

~~where $q_i(\cdot)$ denotes~~ where q_i is the Gaussian reference density with variance ~~$\hat{\sigma}^2 R_{ii}$, with $\hat{\sigma}^2 = \frac{1}{nm-1} \sum_{b \in \mathrm{vec}(\boldsymbol{B})} (b - \bar{b})^2$ and $\bar{b} = \frac{1}{nm} \sum_{b \in \mathrm{vec}(\boldsymbol{B})} b$~~ $\sigma^2 R_{ii}$, evaluated in closed form.

3. Learning the Distance ~~Estimation Using Computer Vision Models~~from a Plot

~~The distance measure defined in Equation ?? quantifies the difference between the true residual distribution P and the reference distribution Q. However, it can only be computed~~ Section 2.2 defined the distance D and showed how to calculate it when the data-generating process ~~is known, which is rarely the case in practice~~for $\boldsymbol{y}$ is known. In practice, this process is unknown, and the analyst has only a fitted model and its residual plot. The distance D is a property of the underlying distribution P, whereas the residual plot is an observed sample that provides partial information about P. The challenge is therefore to predict D from the patterns observed in the plot without explicitly knowing P.

~~To estimate this distance from a residual plot, we~~ We approach this task as a regression problem, training a neural network to predict D from residual plots. Training data are generated through simulation, where the data-generating processes for $\boldsymbol{y}$ are known and the corresponding values of D can be calculated as described in Section 2.2. These values provide numerical training targets, avoiding the need for subjective human assessments. The following subsections describe the network inputs and outputs (Section 3.1), the generation of training data (Section 3.2), and the network itself with the procedure used to train it (Section 3.3).

3.1. How the distance is predicted

We define a computer vision ~~estimator~~

~~$\hat{D} = f_{CV}(V_{h \times w}(\boldsymbol{e}, \hat{\boldsymbol{y}}), \boldsymbol{s}),$~~

model $f_{\boldsymbol{\theta}}$ that maps a residual plot to a non-negative number,

$$\hat{D} = f_{\boldsymbol{\theta}}(V_{h \times w}(\hat{\boldsymbol{e}}, \hat{\boldsymbol{y}}), \boldsymbol{s}), \tag{5}$$

where $V_{h\times w}(\cdot)$ ~~denotes the rendering function that coverts a residuals vs.~~ renders a residuals versus fitted values plot ~~into~~ as an $h \times w$ ~~RGB image, with h and w denoting the image height and width (in pixels), respectively. Here,~~ image, $\boldsymbol{s}$ is a vector of auxiliary features~~including scagnostics measures and sample size, and $f_{CV}(\cdot)$ maps the image and auxiliary features to an estimated distance $\hat{D} \in [0, +\infty)$. Details of the model architecture and training procedure are provided in Sections 3.3 and ??, respectively.~~, and $\boldsymbol{\theta}$ collects the parameters of the network. The image has $h \times w$ pixels with $h, w \in \{32, 64, 128\}$, which allows the effect of resolution on predictive performance to be studied. Points are drawn in black on a white background, so the image is converted to grayscale using the Rec. 601 luma formula (Series, 2011). The conversion preserves the red reference line at $y = 0$.

The auxiliary ~~feature~~ vector $\boldsymbol{s}$ ~~is included to complement information encoded in the residual plot image~~ holds four Scagnostics measures and the sample size. Sample size is included because it cannot be recovered reliably from the image alone. Although residual plots from different sample sizes may exhibit similar patterns, the amount of visual evidence supporting those patterns differs. Moreover, since D grows approximately logarithmically with sample size, providing n explicitly helps the model account for this dependence. The four measures, Monotonic, Sparse, Splines and Striped, are obtained from `cassowaryr` (Mason et al., 2022) and describe the shape, the clustering and the outlying structure of the point cloud. The remaining Scagnostics were excluded because computing the full set was too expensive at inference time.

The predicted distance $\hat{D}$ should be distinguished from the underlying distributional distance D. Different samples drawn from the same distribution P can produce visually different residual plots and hence different predictions, particularly when n is small. Conversely, different distributions can produce visually similar residual plots and hence similar predictions. The model learns to predict the conditional mean of D associated with the visual patterns in the observed plot, rather than inferring the underlying distribution P and then calculating its distance from Q. ~~While $V_{h\times w}(\cdot)$ captures visual patternsof misspecification, certain attributes such as sample sizeare not recoverable from the image. In addition, we include scagnostics as useful graph-based summaries of scatter plotstructure, which have been shown to be effective descriptors of shape, clustering, and outliers in exploratory data analysis.~~

3.2. How the data is generated

A training sample consists of a residual plot and its corresponding value of D. We generate these samples through a single simulation process adapted from Li et al. (2024), which was originally used to produce residual plots for human subject experiments. The process accommodates three types of departure from the classical assumptions: non-linearity in the mean, non-constant variance, and non-Gaussian errors. These departures are controlled by parameters that allow them to occur individually or in combination. Responses are drawn from

$$\begin{aligned}
\boldsymbol{y} &= \mathbf{1}_n + \boldsymbol{x}_1 + \beta_1 \boldsymbol{x}_2 + \beta_2(\boldsymbol{z} + \beta_1 \boldsymbol{w}) + \boldsymbol{k} \odot \boldsymbol{\varepsilon}, \\
\boldsymbol{z} &= \mathrm{He}_j(g(\boldsymbol{x}_1, 2)), \qquad \boldsymbol{w} = \mathrm{He}_j(g(\boldsymbol{x}_2, 2)), \\
\boldsymbol{k} &= \left[\mathbf{1}_n + b(2 - |a|)(\boldsymbol{x}_1 + \beta_1 \boldsymbol{x}_2 - a\mathbf{1}_n)^{\circ 2}\right]^{\circ 1/2},
\end{aligned}$$

where $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are independent predictors, $\mathrm{He}_j(\cdot)$ is the jth-order probabilist's Hermite polynomial (Hermite, 1864), $\odot$ is the Hadamard product, $(\cdot)^{\circ 2}$ and $(\cdot)^{\circ 1/2}$ are the Hadamard square and square root, and

$$g(\boldsymbol{x}, k) = 2k \, \frac{\boldsymbol{x} - x_{\min}\mathbf{1}_n}{x_{\max} - x_{\min}} - k\mathbf{1}_n$$

linearly rescales $\boldsymbol{x}$ so that its minimum and maximum entries, $x_{\min}$ and $x_{\max}$, map to $-k$ and k, respectively, and its intermediate values proportionally.

The ~~distance estimator $\hat{D}$ quantifies how closely the residuals resemble the reference distribution Q and~~ fitted model regresses $\boldsymbol{y}$ on $\boldsymbol{x}_1$, and includes $\boldsymbol{x}_2$ when $\beta_1 \neq 0$. The departure from the fitted model arises from features of the data-generating process that the regression fails to capture. When $\beta_2 \neq 0$, the polynomial terms are omitted, introducing curvature into the residuals. Its shape is determined by j, which counts the turning points of He_j (Figure 3A). The multiplier $\boldsymbol{k}$ makes the error variance a quadratic function of the linear predictor, with a determining where the variance is smallest (Figure 3B) and b controlling its strength (Figure 3C). When a is near -1, the resulting pattern resembles the left-pointing triangle in Figure 1, while values near 1 produce its mirror image. The errors and predictors are drawn from discrete, uniform, normal, or lognormal distributions (Figures 3D and E). The factors listed in Table 1 are fully crossed, ensuring that the training samples cover every combination. Figure 4 illustrates several such combinations.

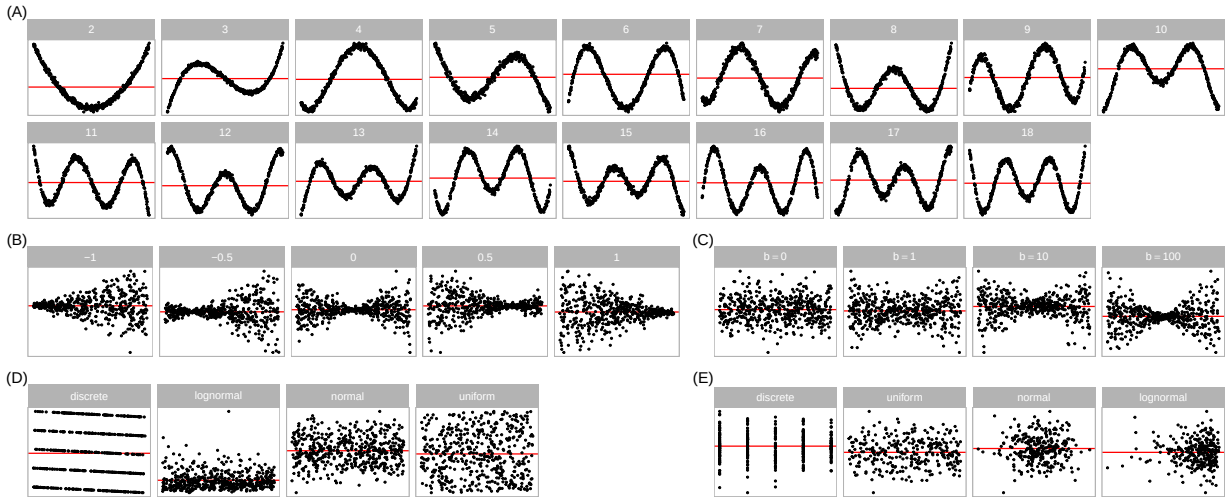


Figure 3: Non-linearity, heteroskedasticity and non-normality forms generated for the synthetic data simulation. Panel (A) shows the 17 shapes produced as the order j of the Hermite polynomial increases. Panel (B) shows the variance patterns produced as the constant a varies from -1 to 1, with the left and right triangles at the extremes. Panel (C) varies the variance multiplier b with a held at zero, from $b = 0$, where the error variance is constant, to $b = 100$. Panel (D) shows the four error distributions, discrete, lognormal, normal and uniform. Panel (E) holds the model fixed and varies only the predictor distribution. The settings are chosen to make each pattern visible, with a small error standard deviation of 0.05 in panel (A) and a large variance multiplier of $b = 100$ in panels (B). The training and test samples carry far more noise than these illustrations, so the same departures are correspondingly harder to see there.

Each training label is computed in one of two ways. When $\mathrm{dist}_\varepsilon$ is normal, D is computed from Equations 1 and 3. Otherwise D is computed from Equations 1 and 4.

We generated 80,000 training samples and 8,000 test samples. To ensure that the model is exposed to the full range of the target, we first randomly selected combinations of factors to generate a large preliminary sample, which was used to estimate the quantiles of D and define 50 bins of equal probability. We then iteratively generated training samples by randomly selecting combinations of factors that were least represented in the current dataset. Each sample was accepted only if its corresponding bin had not yet reached its quota. This process continued until all bins were full, promoting balanced representation across the range of D while ensuring coverage of all factor combinations. Table 2 reports the number of training and test samples for each violation scenario, including those with composite violations. ~~Residual plots generated from the same distribution P may also vary visually, particularly for small n, introducing estimation error that depends on the representativeness of the input plot.~~

3.3. Model ~~Architecture~~architecture and training

The ~~architecture of the computer vision model f_{CV}~~ network is adapted from the well-established VGG16 architecture (Simonyan and Zisserman, 2014). ~~While~~ Although more recent architectures ~~like~~ such as ResNet

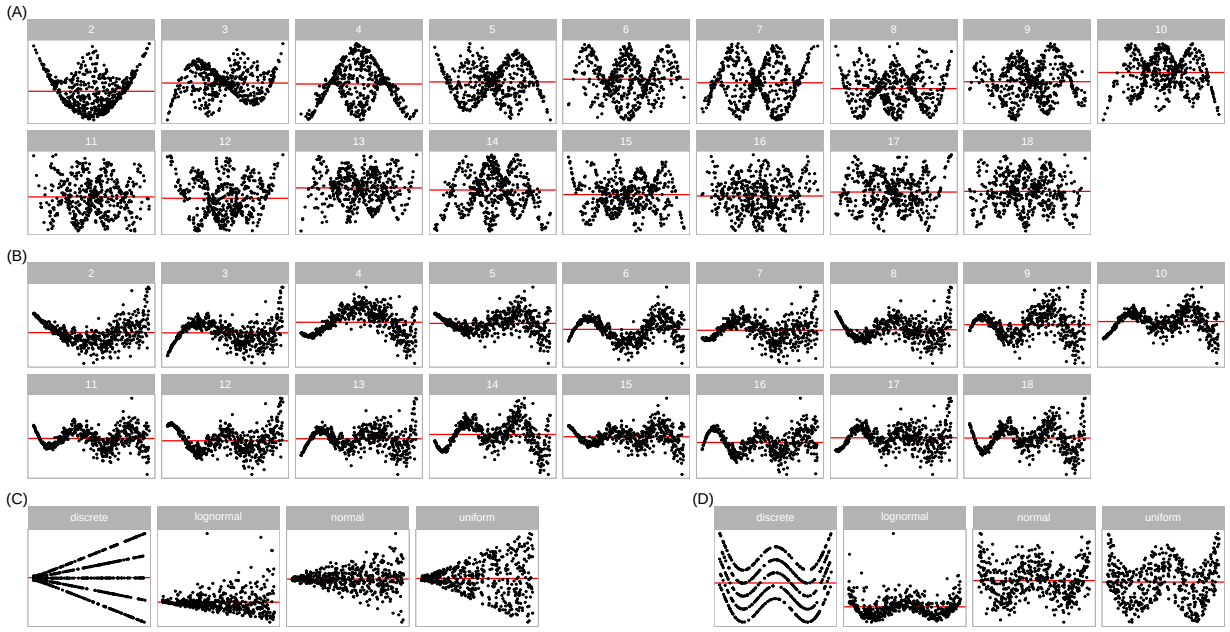


Figure 4: The training samples cover every combination of the factors in Table 1, and these panels show a few of them. Panel (A) repeats the 17 shapes of Figure 3A with a second predictor in the simulation. Panel (B) combines the non-linearity with heteroskedasticity. Panel (C) combines the four error distributions with heteroskedasticity. Panel (D) combines the non-linearity with the four error distributions.

Table 1: Factors in the synthetic data generating process and their domains. The order j controls the shape of the non-linearity, a the shape of the heteroskedasticity, b, σ_ε and n the signal strength, and the three distributions the shape of the predictors and the errors.

Factor	
j	
a	
b	
β_1	
β_2	
dist_ε, dist_{x_1},	~~after adjusting for sample size effects, can be used as an index of model violation severity (see Appendix D~~
σ_ε	
σ_{x_1}	
σ_{x_2}	
n	

(He et al., 2016) and DenseNet (Huang et al., 2017) ~~, have achieved even greater performance~~have achieved higher accuracy on image classification benchmarks, VGG16 ~~remains a solid choice for many applications due to its simplicity and effectiveness. Our decision to use VGG16 aligns with our goal of starting with a~~ was chosen as a proven and straightforward ~~model~~architecture with a simple yet effective block structure. Figure 5 ~~illustrates the architecture, while Appendix B provides detailed layer specifications~~shows the layout.

The model ~~accepts residual plots rendered as $h \times w$ RGB images, with $h, w \in \{32, 64, 128\}$, allowing us to study the effect of image resolution on predictive performance while balancing computational cost. Images are first converted to grayscale using the Rec. 601 luma formula (Series, 2011), as the data points are plotted in black on a white background. The background also includes a red reference line at $y = 0$,~~

Table 2: Number of training and test samples for each violation scenario, including cases with multiple simultaneous violations. Each sample consists of a residual plot as input and the corresponding distance D as the target. Root mean square error (RMSE) values for the training and test sets are shown for each scenario.

	Train		Test	
Violations	#samples	RMSE	#samples	RMSE
no violations	1,546	1.149	155	1.267
non-linearity	22,184	0.625	2,218	0.787
heteroskedasticity	10,826	0.528	1,067	0.602
non-linearity + heteroskedasticity	10,221	0.593	985	0.751
non-normality	10,797	0.279	1,111	0.320
non-linearity + non-normality	8,835	0.472	928	0.600
heteroskedasticity + non-normality	7,870	0.354	819	0.489
non-linearity + heteroskedasticity + non-normality	7,721	0.432	717	0.620

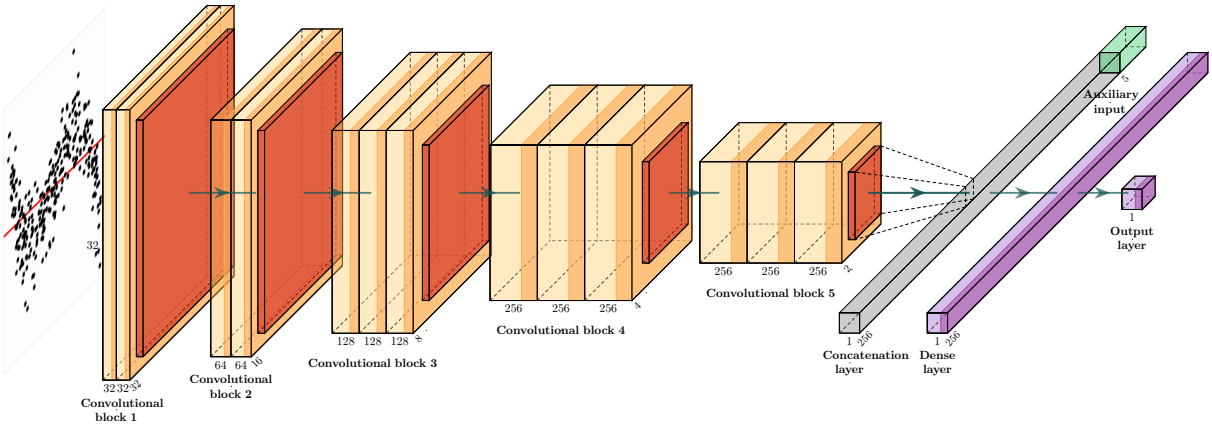


Figure 5: Diagram of the architecture of the optimized computer vision model. Numbers at the bottom of each box show the shape of the output of each layer. The band of each box drawn in a darker color indicates the use of the rectified linear unit activation function. Yellow boxes are 2D convolutional layers, orange boxes are pooling layers, the grey box is the concatenation layer, and the purple boxes are dense layers.

~~which is preserved under this conversion.~~

~~The processed images are used as input to the first convolutional block. The model~~ consists of up to five ~~consecutive~~ convolutional blocks, ~~mirroring the original VGG16 architecture. Within each block~~ , with the number of blocks treated as a hyperparameter. Each block applies two 3×3 ~~convolutional layers are applied~~convolutions, each followed by batch ~~normalization for training stability and a rectified linear unit (ReLU) activationfor non-linearity. The block concludes with~~ normalisation and a ReLU activation, followed by a 2×2 max-pooling layer~~that downsamples the spatial dimensions by taking the maximum over non-overlapping windows for each channel. A dropout layer is then applied at the end of each block to randomly deactivate activations during training, improving regularization~~, then dropout.

The ~~final convolutional output is aggregated~~ output of the final block is reduced to a vector using either global max-pooling or global average-pooling, ~~producing a two-dimensional feature representation. This feature representation~~ with the choice treated as a hyperparameter. This vector is concatenated with ~~an~~ the $n \times 5$ tensor ~~containing the "Monotonic", "Sparse", "Splines", and "Striped" measures computed using the~~

~~`cassowaryr` R package (Mason et al., 2022), along with the sample sizefor each residual plot.~~[1]

~~The concatenated feature tensor is passed to the final prediction block, which consists of two fully connected~~ of auxiliary features, comprising four Scagnostics and the sample size, to combine visual and tabular information before prediction. The fused representation is then passed through two dense layers. The first ~~layer~~ has at least ~~128 unitsand is~~ 128 units, followed by batch ~~normalization and dropoutfor regularization~~normalisation and dropout. The second ~~fully connected layer contains~~ has a single unit ~~, followed by~~ with a ReLU activation to ~~enforce~~ ensure non-negative ~~outputs, producing the final scalar prediction~~predictions.

3.4. ~~Data Generation and Model Training~~

~~To train the computer vision model for estimating $\hat{D}$, we constructed a dataset of paired residual plots and their corresponding D values. In deep learning, synthetic data are commonly used for training vision models when labelled data are scarce, costly, or difficult to obtain reliably (Nikolenko et al., 2021). This is particularly appropriate when the data-generating mechanism is well specified, enabling controlled generation of large-scale examples that encode known structural violations. Accordingly, we generated 80,000 training samples and 8,000 test samples, providing full control over the generative process, exact computation of D, and broad coverage of residual patterns associated with model misspecification.~~

~~The simulated data incorporated three common types of residual departures: non-linearity, heteroskedasticity, and non-normality. These effects were induced by fitting a simple linear regression model to data generated from a more flexible underlying process based on Hermite polynomials (Hermite, 1864), multiple predictors, and a range of predictor and error distributions. A comprehensive grid of parameter settings was used to produce diverse residual behaviours, including configurations with multiple simultaneous violations.~~

~~To ensure uniform coverage across the target space of D, we adopted a bucketed sampling strategy. Specifically, simulated samples were iteratively generated and assigned to one of 50 bins corresponding to quantiles of D, with acceptance controlled to balance representation across difficulty levels. Table 2 summarises the resulting sample sizes for each violation scenario.~~

~~Number of training and test samples for each model violation scenario, including cases with multiple simultaneous violations. Each sample consists of a residual plot as input and the corresponding distance D as the target. Root mean square error (RMSE) values for the training and test sets are shown for each scenario. Details of the synthetic data generating process used to construct these samples are provided in Appendix A.~~

~~Computer vision model parameters $\boldsymbol{\theta}$ were randomly initialized using the Glorot Uniform method (Glorot and Bengio, 2010) and optimized with the Adam optimizer (Kingma and Ba, 2014) under~~ Parameters are initialised with the Glorot uniform scheme (Glorot and Bengio, 2010) and optimised with Adam (Kingma and Ba, 2014) against a mean squared error ~~objective:~~ loss,

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} (D_i - f_{\boldsymbol{\theta}}(V_i, \boldsymbol{s}_i))^2,$$

where n_{train} ~~denotes~~ is the number of training samples~~, V_i is the i-th residual plot, and $\boldsymbol{s}_i$ represents the auxiliary tabular features, including four scagnostics measures and sample size~~.

~~The model was trained on paired inputs consisting of RGB residual plot images and corresponding D values, with the objective of learning the mapping $\hat{D} = f_{\boldsymbol{\theta}}(V, \boldsymbol{s})$. Training was conducted on the MASSIVE M3 high-performance computing platform (Goscinski et al., 2014) using TensorFlow (Abadi et al., 2016) and Keras (Chollet et al., 2015). Hyperparameters were tuned using Bayesian optimisation via~~ Hyperparameters were chosen by Bayesian optimisation with KerasTuner (O'Malley et al., 2019), ~~minimizing~~ minimising the

[1] ~~Note: These measures were selected for their reliability and computational efficiency. Other scagnostics were excluded because they occasionally caused R process crashes ($\sim$5%) during data preparation due to a bug in the `interp` package (Gebhardt et al., 2023). Although this issue was later fixed at our request, the corrected version was not available during model training, and the full set of features was also deemed too computationally expensive for efficient inference.~~

validation root mean square error ~~(RMSE)~~ over 100 trials. The search ~~space included~~ covered the dropout rate, ~~batch normalization configuration, input resolution, and the inclusion of auxiliary features .~~

~~Further details on the synthetic data-generating process, parameter settings, sampling scheme, example residual plots, and hyperparameter search configuration are provided in Appendix A~~ the batch normalisation configuration, the input resolution and ~~Appendix C~~whether the auxiliary features were included. Training ran on the MASSIVE M3 platform (Goscinski et al., 2014) with TensorFlow (Abadi et al., 2016) and Keras (Chollet et al., 2015). Appendix A records the search space and the values selected.

4. Statistical ~~Testing~~testing

Under ~~the null hypothesis~~ H_0, ~~corresponding to a correctly specifiedmodel, the distance satisfies $D = 0$ because $P = Q$. In practice, however, the estimated~~ the model is correctly specified, so $P = Q$, $\mathcal{M}(P) = \mathcal{M}(Q)$, and $D = 0$. However, the predicted distance $\hat{D}$ ~~is generally non-zero, even for null plots, due to sampling variability and imperfections in the computer vision model. Consequently, statistical significance should be assessed~~ of an observed residual plot need not be zero. The plot is a finite sample, and the network is an imperfect predictor of D. Therefore, $\hat{D}$ must be interpreted relative to the ~~distribution of estimated distances obtained from null plots rather than by the magnitude of $\hat{D}$ alone~~distances the same network assigns to plots generated under H_0. We construct a test by comparing the observed plot with null plots that preserve its fitted values and residual norm.

~~To calibrate $\hat{D}$, we adopt the residual rotation procedure used for null plotgeneration in Buja et al. (2009) . This involves drawing an auxiliary response $\boldsymbol{w}$ from $\mathcal{N}(\mathbf{0}_n, \mathbf{I}_n)$ and compute the projected residuals $\hat{\boldsymbol{e}}_w = \boldsymbol{R}\boldsymbol{w}$. These are then rescaled to match the Euclidean norm of the observed residuals, $\tilde{\boldsymbol{e}}_w = \hat{\boldsymbol{e}}_w \cdot \frac{\|\hat{\boldsymbol{e}}\|}{\|\hat{\boldsymbol{e}}_w\|}$, so that $\|\tilde{\boldsymbol{e}}_w\| = \|\hat{\boldsymbol{e}}\|$. A null plotis then the scatter plot of $(\hat{\boldsymbol{y}}, \tilde{\boldsymbol{e}}_w)$, which under the null hypothesis of a correctly specified model is statistically indistinguishable from the true residual plot $(\hat{\boldsymbol{y}}, \hat{\boldsymbol{e}})$.~~

4.1. The test

Let $\hat{D}$ denote the ~~estimated distance for the true residual plot and~~ predicted distance of the observed residual plot, and let ~~$\{\hat{D}^{(i)}_{null}\}_{i=1}^m$~~ $\{\hat{D}^{(i)}_{\text{null}}\}_{i=1}^m$ denote the ~~estimated distances for~~ predicted distances of m ~~null plots generated under H_0. For a lineup containing $m + 1$ plots (one true plot and m null plots),~~ null plots. We calculate the Monte Carlo p-value

$$\hat{p} = \frac{1}{m+1}\left(1 + \sum_{i=1}^{m} \mathbb{I}\{\hat{D}^{(i)}_{\text{null}} \geq \hat{D}\}\right).$$

The level-α test rejects H_0 when $\hat{p} \leq \alpha$, that is, when $\hat{D}$ exceeds the $(1-\alpha)$ quantile $Q_{\text{null}}(1-\alpha)$ of the null distances. The observed distance is thus assessed against the null distances rather than against zero, allowing for the network's prediction error without requiring its predictions to be unbiased or accurate for individual plots.

The lineup protocol is a special case of this test. When $\alpha = 1/(m+1)$, rejection requires

$$\hat{D} > \max_{1 \leq i \leq m} \hat{D}^{(i)}_{\text{null}}.$$

The observed plot must have a strictly greater predicted distance than every null plot. This is the ~~visual inference decision rule is~~ decision rule of a lineup with m null plots and one observed plot: the null is rejected only when the viewer identifies the observed plot as the most unusual. With the standard lineup size of $m = 19$, this corresponds to a 5% test.

4.2. Rotation and its exactness

We generate null plots by rotating the observed residual vector while preserving its norm and the fitted values, following the rotation construction of Buja et al. (2009). For each null plot, we draw

$$\boldsymbol{w} \sim \mathcal{N}(\mathbf{0}_n, \boldsymbol{I}_n)$$

and project it onto the residual subspace:

~~Reject $H_0 \iff > \max_{1 \le i \le m} \hat{D}_{null}^{(i)}$~~ $\hat{\boldsymbol{e}}_w = \boldsymbol{R}\boldsymbol{w}$.

~~This corresponds to the standard lineup protocol, where the true residual plot must be identified as the most visually distinct plot~~We then rescale the projected vector to match the observed residual norm:

$$\tilde{\boldsymbol{e}}_w = \hat{\boldsymbol{e}}_w \frac{\|\hat{\boldsymbol{e}}\|}{\|\hat{\boldsymbol{e}}_w\|},$$

so that $\|\tilde{\boldsymbol{e}}_w\| = \|\hat{\boldsymbol{e}}\|$. The resulting null plot is the scatter plot of $(\hat{\boldsymbol{y}}, \tilde{\boldsymbol{e}}_w)$. Thus, each null plot preserves the fitted values and residual norm of the observed plot, differing only in the direction of its residual vector.

~~More generally, the empirical distribution of $\{\hat{D}_{null}^{(i)}\}_{i=1}^{m}$ provides an estimate of the null distribution~~ Under H_0, the fitted values $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$ and the residuals $\hat{\boldsymbol{e}} = \boldsymbol{R}\boldsymbol{y}$ are independent, since $\mathrm{Cov}(\hat{\boldsymbol{y}}, \hat{\boldsymbol{e}}) = \sigma^2 \boldsymbol{H}\boldsymbol{R} = \mathbf{0}$ and both are Gaussian. The direction of $\hat{\boldsymbol{e}}$ is uniform on the unit sphere of the residual subspace and independent of its norm. Since the projected Gaussian vector $\boldsymbol{R}\boldsymbol{w}$ has the same isotropic Gaussian distribution within the residual subspace, rescaling it to the observed residual norm gives a direction with the same conditional distribution as the observed residual direction, given the fitted values and residual norm.

The following proposition uses this property to establish the finite-sample Type I error guarantee of the test~~statistic.~~.

We define a scoring function as any fixed map from the collection of pairs $\{(\hat{y}_i, \hat{e}_i)\}_{i=1}^{n}$ to a real number. The trained model $f_{\boldsymbol{\theta}}$, together with a fixed plotting and rendering pipeline that specifies the aesthetic mappings, layout, and pixel resolution, induces one such function. Any auxiliary features are derived from the same pairs, so the score remains a function of the collection. The result below holds for any fixed scoring function, regardless of how accurately it predicts D.

Proposition 1. *Under H_0, for any fixed scoring function f and any $\alpha \in (0, 1)$, the test of Section 4.1 satisfies*

$$\Pr(\hat{p} \le \alpha \mid H_0) \le \frac{\lfloor \alpha(m+1) \rfloor}{m+1} \le \alpha.$$

If the $m+1$ predicted distances are almost surely distinct, the first inequality is an equality.

Proof. Let ~~$Q_{null}(\tau)$~~ $C = (\hat{\boldsymbol{y}}, \|\hat{\boldsymbol{e}}\|)$ denote the fitted values and the norm of the observed residual vector. Under H_0, conditional on C, the observed residual direction and the independently generated rotated residual directions are independent draws from the same uniform distribution on the unit sphere of the residual subspace. Hence, the observed residual vector and the m rotated residual vectors are exchangeable. Because C is common to all plots, applying the same plotting procedure and fixed scoring function f preserves this exchangeability. Thus, the resulting $m+1$ predicted distances are exchangeable conditional on C.

Let K denote the ~~empirical τ-quantile of this distribution. A level-α test is then given by~~

~~Reject $H_0 \iff \hat{D} \ge Q_{null}(1-\alpha)$.~~

rank of the observed distance among the $m+1$ distances, counting from the largest and breaking ties uniformly at random, independently of the data. By conditional exchangeability, K is uniformly distributed

on $\{1, \ldots, m+1\}$ given C under H_0. Each of the $K-1$ null distances ranked above the observed distance is at least $\hat{D}$, so including the observed distance itself in the upper-tail count gives

$$\hat{p} \geq \frac{K}{m+1},$$

~~Equivalently, a Monte Carlo p-value can be computed as~~

~~$$\hat{p} = \frac{1}{m+1} + \frac{1}{m+1} \sum_{i=1}^{m} \mathbb{I}\left(\hat{D}_{null}^{(i)} \geq \hat{D}\right).$$~~

with equality when the observed distance is distinct from all m null distances. Since $\hat{p} \geq K/(m+1)$,

$$\begin{aligned} \Pr(\hat{p} \leq \alpha \mid H_0, C) &\leq \Pr\left(K \leq \alpha(m+1) \mid H_0, C\right) \\ &= \Pr\left(K \leq \lfloor \alpha(m+1) \rfloor \mid H_0, C\right) \\ &= \frac{\lfloor \alpha(m+1) \rfloor}{m+1} \\ &\leq \alpha. \end{aligned}$$

The second equality holds because K takes integer values, so $\Pr(K \leq x) = \Pr(K \leq \lfloor x \rfloor)$. The third follows from the conditional uniform distribution of K. The final inequality follows from $\lfloor x \rfloor \leq x$. Taking expectations with respect to the distribution of C under H_0 gives

$$\Pr(\hat{p} \leq \alpha \mid H_0) \leq \frac{\lfloor \alpha(m+1) \rfloor}{m+1} \leq \alpha.$$

~~This is the standard simulation-based estimator of the tail probability under H_0. For computational efficiency, quantiles of the null distribution may be pre-computed over a range of sample sizes and stored in a lookup table, allowing approximate~~ If the $m+1$ predicted distances are almost surely distinct, then $\hat{p} = K/(m+1)$, so the first inequality is an equality. □

The test therefore provides finite-sample Type I error control, with attainable p-values ~~to be obtained without generating new null plots.~~ in increments of $1/(m+1)$. For $m = 19$, the smallest attainable p-value is $1/20 = 0.05$, corresponding to the lineup decision rule. When the predicted distances are almost surely distinct, the test attains the corresponding attainable significance levels. Ties can make the test conservative but do not invalidate the guarantee.

5. ~~Bootstrap Variability Assessment~~

The validity of the test follows from the rotation-based exchangeability argument, not from the predictive accuracy of the network. Any fixed scoring function applied consistently to the observed and null plots yields the same finite-sample guarantee. The choice of scoring function instead affects power. A network that ranks departures from the null more effectively can better distinguish an observed plot from its null counterparts.

~~Bootstrap methods are commonly used in linear regression to assess parameter variability without strong distributional assumptions (Davison and Hinkley, 1997; Efron and Tibshirani, 1994). This involves resampling the observed data (pairs of responses and predictors) with replacement and refitting the model on each bootstrap sample.~~

4.1. Bootstrap variability

~~Similarly, this case-resampling bootstrap can be used to assess the variability of the estimated distance $\hat{D}$. For each bootstrap sample $i = 1, \ldots, n_{boot}$, a regression model $M^{(i)}_{boot}$ is refitted using the resampled data, producing a residual plot $V^{(i)}_{boot}$ and a corresponding estimated distance $\hat{D}^{(i)}_{boot}$.~~ Refitting the model on resampled data provides a complementary view of the variability of the same statistic. For each of the n_{boot} case-resampling bootstrap samples, we refit the model, render a residual plot, and predict its distance. Each refitted model also has its own null distribution, from which a p-value is calculated. The ~~empirical distribution of $\{\hat{D}^{(i)}_{boot}\}_{i=1}^{n_{boot}}$ provides an estimate of the sampling distribution of $\hat{D}$, from which confidence intervals may be constructed.~~

~~More generally, each bootstrap model $M^{(i)}_{boot}$ induces its own null distribution of estimated distances . Let $Q^{(i)}_{boot}(\tau)$ denote the τ-quantile of the corresponding null distribution. A level-α test for the i-th bootstrap model is then given by~~

$$\text{~~Reject } H_0 \iff \hat{D}^{(i)}_{boot} \geq Q^{(i)}_{boot}(1-\alpha).$$

~~The~~ proportion of bootstrap ~~models for which H_0 is rejected,~~

$$\frac{1}{n_{boot}} \sum_{i=1}^{n_{boot}} \mathbb{I}\left(\hat{D}^{(i)}_{boot} \geq Q^{(i)}_{boot}(1-\alpha)\right),$$

samples in which the test rejects the null hypothesis estimates how often the ~~fitted regression~~ model would be judged misspecified under repeated sampling~~from the same data-generating process.~~

. This procedure is computationally expensive~~because it requires evaluating $n_{boot}(m+1)$ residual plots , where~~ , requiring $n_{\text{boot}}(m+1)$ residual plots in total, as each refit contributes one observed plot and m ~~denotes the number of null plots~~per bootstrap sample~~~~null plots. In practice, ~~the bootstrap-specific critical values $Q^{(i)}_{boot}(1-\alpha)$ may be approximated by the corresponding quantile $Q_{null}(1-\alpha)$~~ each refitted model can instead be compared against the null plots generated from the original ~~null distribution, substantially reducing the computational burden~~model, reducing the total to n_{boot} residual plots.

5. Results

5.1. Model ~~Performance~~performance

Table 3 summarizes the test performance of the optimized models across three input sizes, using 80,000 training samples and 8,000 test samples generated as described in Section ~~??~~3.2. The 32×32 model consistently outperformed the others, with a mean absolute error of approximately 0.43, a negligible deviation given the typical D range of 0 to 7. High R^2 values also indicated strong linear correlation between predictions and targets. Based on these results, we used the best-performing 32×32 model for subsequent analyses.

Figure 6A shows a hexagonal heatmap of $D - \hat{D}$ versus D. Smoothing curves (fitted by generalized additive models, Hastie, 2017) reveal that all models performed well for $1.5 < D < 6$, where no structural issues were observed. Over-prediction occurred when $D < 1.5$, and under-prediction primarily when $\hat{D} > 6$.

For null plots ($D = 0$), over-prediction was expected (see Section 4), but not under-prediction. Therefore, we analysed the relationship between residuals and all the factors involved in the data generating process. We found that most deviations stemmed from non-linearity and the presence of a second predictor in the synthetic data model, where small error variances caused under-predictions and large error variances caused over-prediction, as illustrated in Figure ~~??~~6B.

To investigate further, we stratified the ~~train and test set~~ training and test sets by violation type and re-evaluated model performance (Table 2). ~~It was found that null~~ Null plots had the ~~poorest metrics, while~~ highest prediction errors, whereas non-normality ~~cases were~~ was the easiest to predict, with ~~RMSE around~~ an RMSE of approximately 0.3. Non-linearity ~~(RMSE $\approx$ 0.8) was harder to assess~~ was more difficult to assess (RMSE $\approx$ 0.8) than heteroskedasticity (~~RMSE $\approx$ 0.6~~RMSE $\approx$ 0.6) or non-normality~~(RMSE $\approx$ 0.3). Plots with~~ . Plots containing multiple violations showed intermediate performance.

Table 3: ~~The test~~ Test performance of three optimized models with different input sizes. ~~The metrics are computed by comparing~~ Metrics compare the ~~estimated~~ predicted distance $\hat{D}$ (model output) ~~and~~ with the target distance D generated from the synthetic data model. RMSE ~~represents~~ is the root mean squared error~~;~~, R^2 ~~is~~ the squared correlation between the two~~quantities;~~, MAE ~~denotes~~ the mean absolute error~~;~~, and Huber loss ~~refers to the~~ its average ~~Huber loss computed~~ over the test set.

	RMSE	R^2	MAE	Huber loss
32×32	0.660	0.901	0.434	0.18
64×64	0.674	0.897	0.438	0.19
128×128	0.692	0.892	0.460	0.20

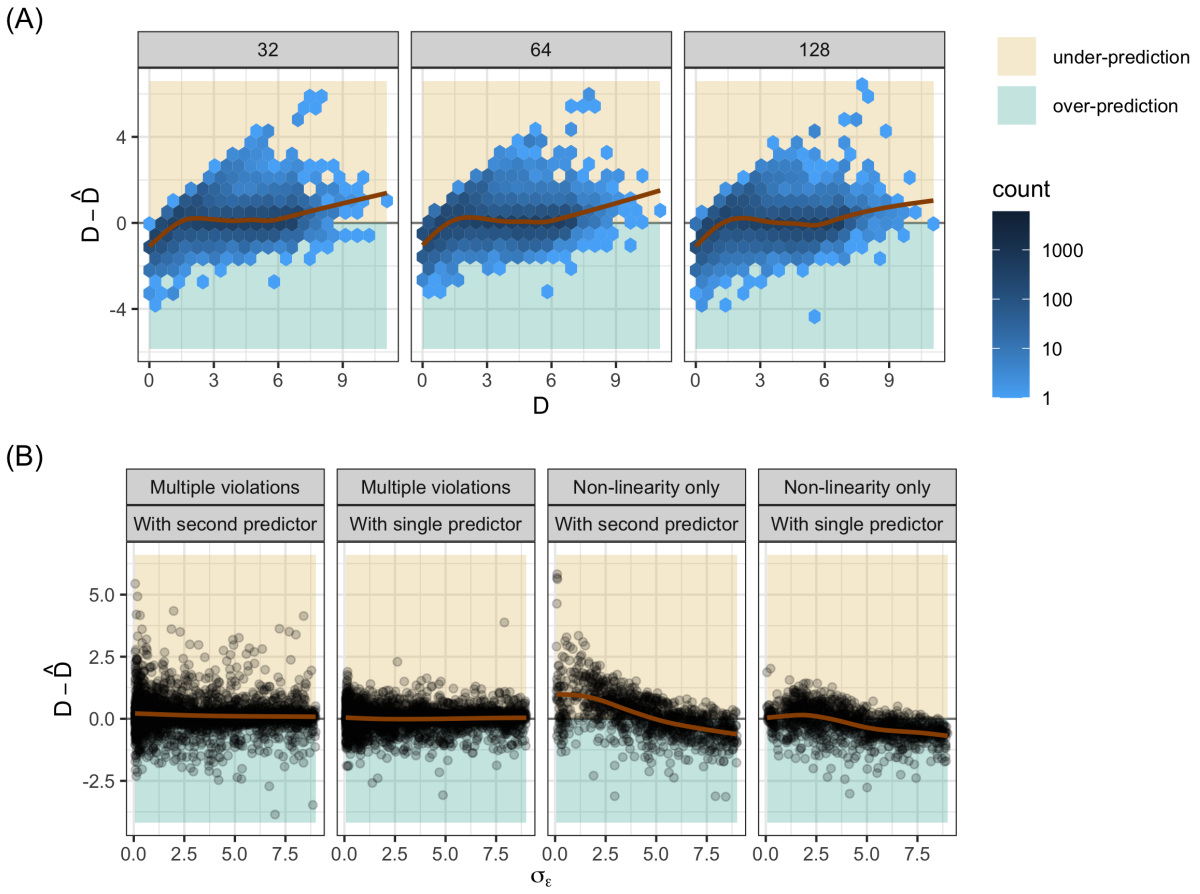


Figure 6: ~~Hexagonal heatmap for difference in~~ Panel (A) shows hexagonal heatmaps of the prediction error $\hat{D} - D$ against D ~~and $\hat{D}$ vs D~~ on test data for three optimized models with different input sizes. Panel (B) shows scatter plots of $\hat{D} - D$ against σ_ε on test data for the 32×32 optimized model, grouped by whether the regression has only a non-linearity violation and whether it includes a second predictor. The brown lines are smoothing curves produced by fitting generalized additive models. ~~Over-prediction and under-prediction can be observed~~ The model over-predicts for small D and under-predicts for large D~~respectively~~, and it over-predicts for small σ_ε when the regression has a second predictor.

5.2. *Comparison with human visual inference and conventional tests*

Evaluating a method that automates visual inference requires a benchmark for determining which departures from model assumptions are meaningful in practice. Li et al. (2024) provided relevant experimental

evidence by comparing human judgements with conventional residual-based tests using lineups of residual plots. The study found that conventional tests judged regression models inadequate almost twice as often as human viewers. Many of these rejections arose from departures that were not visually distinguishable from null plots and were too small to meaningfully affect downstream analyses. Although conventional tests can detect small but real departures, these findings suggest that statistical significance alone may not reflect the practical importance of a departure. The lineups from this experiment provide human assessments of individual plots, allowing us to evaluate how closely our method's resulting decisions correspond to human judgements.

~~Scatter plots for difference in D and $\hat{D}$ vs σ on test data for the 32×32 optimized model. The data is grouped by whether the regression has only non-linearity violation, and whether it includes a second predictor in the regression formula. The brown lines are smoothing curves produced by fitting generalized additive models. Under-prediction mainly occurs when the data-generating process has small σ, a second predictor, and only non-linearity as the model violation.~~

5.3. ~~*Comparison with Human Visual Inference and Conventional Tests*~~

5.2.1. ~~*Overview of the Human Subject Experiment*~~

5.2.1. *Overview of the human subject experiment*

~~In order to check the validity of the proposed computer vision model , residual plotspresented in the human subject experiment conducted by Li et al. (2024) was assessed.~~ The study collected ~~7,974~~ 8,860 responses across 1,~~152~~ 176 lineups, each containing one randomly positioned ~~true~~ observed residual plot and 19 null plots. ~~Of these, 24 were~~ Excluding attention-check lineups ~~with highly obvious patterns, 36 were null~~ and lineups containing only null plots ~~, 279 involved uniformly distributed predictors and were evaluated by 11 participants each,and the remaining 813 involved discrete, skewed, or normally distributed predictors and were evaluated by 5 participants each. Attention-check and null lineups are excluded from the present~~ leaves 1,116 lineups and 7,254 responses in the analysis.

Participants were recruited through Prolific. Most were not experts in residual plots, and 56.7% reported no prior experience with studies involving data visualisations. ~~The sample was approximately gender-balanced, more than half held a diploma or bachelor's degree, and almost all participants were between 18 and 39 years of age.~~ Consequently, the results primarily reflect the ability of non-experts to distinguish true residual plots from null plots based on visual differences rather than specialised statistical training.

The residual plots in Li et al. (2024) were generated from a special case of the synthetic data model ~~used in this study~~of Section 3.2. Model violations were introduced independently, so non-linearity and heteroskedasticity did not co-occur within the same lineup. In addition, the experimental design did not consider non-normality or multiple predictors.

5.2.2. ~~*Model*~~ *Performance on the* ~~*Human-evaluated Data*~~*human-evaluated lineups*

~~Each lineupin Li et al. (2024) contains one true residual plot and 19 null plots. While the true residual plot's distance~~ For each lineup, we predicted D ~~depends on the data-generating process, null plots have $D = 0$. We used our optimized computer vision model to estimate D~~ for all plots ~~. To ensure fair comparison, we reject~~ and rejected H_0 ~~if the true residual plot has the highest estimated distance. Conventional tests , including~~ when the observed plot had the largest predicted distance, which at 19 nulls is the 5% level the conventional tests use. Each conventional test was applied to the lineups whose violation it is designed to detect, the Ramsey RESET (Ramsey, 1969) ~~for~~ to those carrying non-linearity and the ~~Breusch-Pagan test (Breusch and Pagan, 1979) for heteroskedasticity, were also applied to the same data~~Breusch–Pagan test (Breusch and Pagan, 1979) to those carrying heteroskedasticity.

Table 4 reports performance metrics for $\hat{D}$ on ~~true residual plots~~the observed residual plot of each lineup, by violation type. While the performance is slightly worse than on the test data provided in Table 3, the mean absolute error remains low at 0.56 and the correlation between predictions and true values remains high at 0.90. Consistent with results in Table 2, non-linearity cases are harder to predict than heteroskedasticity cases.

Table 4: The performance of the 32×32 model on the data used in the human subject experiment.

Violation	RMSE	R^2	MAE	Huber loss
heteroskedasticity	0.721	0.852	0.553	0.235
non-linearity	0.738	0.770	0.566	0.246

~~Summary of the comparison of decisions made by computer vision model with decisions made by conventional tests and visual tests conducted by human.~~

Table 5: Agreement between the decisions of the computer vision method, conventional tests, and human visual tests on the same lineups.

Violations	#Samples	#Agreements	Agreement rate
Compared with conventional tests			
heteroskedasticity	540	~~464~~ 429	~~0.8593~~0.7944
non-linearity	576	~~459~~ 427	~~0.7969~~0.7413
Compared with visual tests conducted by human			
heteroskedasticity	540	~~367~~ 394	~~0.6796~~0.7296
non-linearity	576	~~385~~ 419	~~0.6684~~0.7274

Table 5 compares the ~~model~~computer vision method's rejection decisions with those of conventional tests and human visual tests. ~~Agreement rates are 85.95~~The agreement rates with conventional tests are 79.44% for heteroskedasticity cases and ~~79.69~~74.13% for non-linearity cases, ~~higher than those observed for~~ while those with human visual tests ~~(67.96% and 66.84~~are 72.96% and 72.74%, respectively~~). This suggests that the model aligns~~ . The method thus aligns with the two benchmarks to a similar degree, slightly more closely with ~~the best available conventional tests . However, as Figure ?? shows, the model does not always reject when conventional tests do, indicating a lower sensitivity to model violations~~conventional tests than with human visual judgements, but the moderate level of agreement indicates that it does not fully reproduce either.

~~Rejection rate (p-value ≤ 0.05) of computer vision models conditional on conventional tests on non-linearity (left) and heteroskedasticity (right) lineups displayed using a mosaic plot. When the conventional test fails to reject, the computer vision mostly fails to reject the same plot as well as indicated by the height of the top right yellow rectangle, but there are non negligible amount of plots where the conventional test rejects but the computer vision model fails to reject as indicated by the height of the top left yellow rectangle.~~

~~To further illustrate these patterns, Figure ?? presents parallel coordinate plots comparing~~ Figure 7 shows a parallel coordinate plot of the decisions from human visual tests, the computer vision ~~model~~method, and conventional tests for the same lineups. All three ~~agree in about 50% of cases. When~~ procedures agree in just over half of the lineups, rejecting 32% of them together and accepting 24% together, and disagree on the remaining 44%. Most of the remaining disagreement comes from lineups that humans do not reject but conventional tests do. Among lineups that humans do not reject, conventional tests reject ~~in about half of those cases~~about half, whereas the ~~model rejects in~~ method rejects only 25%. ~~Notably, the model rarely rejects when both humans and~~ Conversely, the method rarely rejects lineups that conventional tests do not, ~~suggesting that while it is more sensitive than humans , its rejection behavior is more aligned with human judgment than with~~ and rarely misses those that humans reject. The method's decisions therefore fall between the two benchmarks. It flags more violations than humans do, but is more conservative than conventional tests.

~~Figure 8 plots rejection decisions against the true distance~~ Among lineups with $|D - 2| \leq 0.5$, the average rejection rates for the computer vision method are 31% for non-linearity lineups and 48% for heteroskedasticity lineups, intermediate between conventional tests (75% and 68%, respectively) and visual

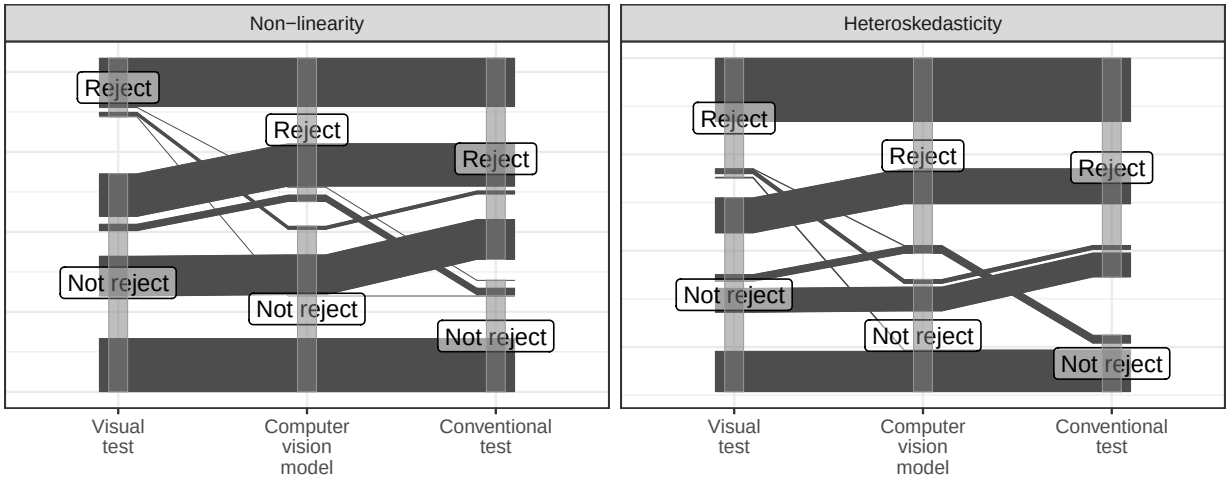


Figure 7: Parallel coordinate plots of the decisions made by the human visual tests, the computer vision model ~~,~~ and the conventional ~~tests~~ test, for non-linearity (left) and ~~visual tests made by human~~heteroskedasticity (right) lineups. All three procedures agree in ~~around 50~~just over half of the lineups, rejecting 32% of ~~cases~~them together and accepting 24% together. ~~The model~~ Among lineups that humans do not reject, the method rejects less often than conventional tests ~~when humans do~~not, and rarely rejects when both others do not, indicating closer alignment with human judgment.

tests (3% and 18%). As D ~~. Compared to conventional tests, the computer vision model rejects fewer cases when $D < 4$, but a similar number of cases when $D > 4$. This indicates that the model~~ increases, the differences narrow. Among lineups with $|D - 4| < 0.5$, the method's average rejection rates rise to 90% and 92%, close to the rates for conventional tests (93% and 97%). Visual tests remain less sensitive, reaching near-universal rejection only beyond $D = 5$, compared with $D = 4$ for both the method and conventional tests. Overall, the method is less sensitive to minor ~~deviations~~ departures from model assumptions ~~, while remaining comparably sensitive to more substantial violations. Visual tests are the least sensitive overall, showing few rejections for $D < 2$, and requiring a higher threshold ($D > 5$) for near-universal rejection , compared to $D > 4$ for both the model and conventional tests~~ than conventional tests, while retaining comparable sensitivity to substantial departures, placing its rejection behaviour between that of conventional tests and visual assessment.

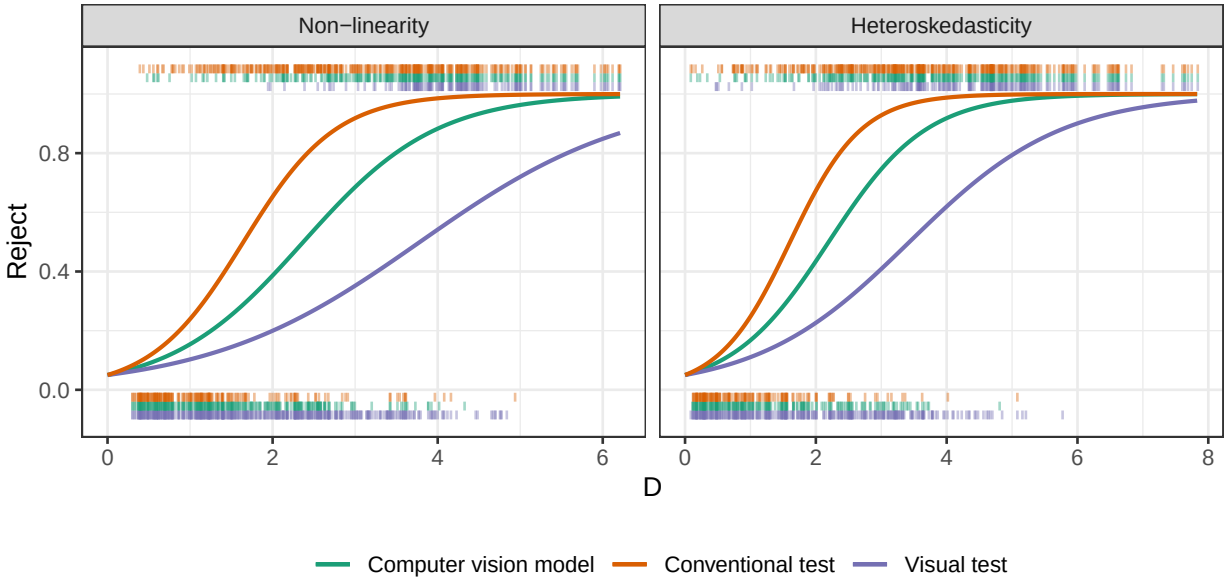
~~Comparison of power of visual tests, conventional tests and the computer vision model. Marks along the x-axis at the bottom of the plot represent rejections made by each type of test. Marks at the top of the plot represent acceptances. Power curves are fitted by logistic regression models with no intercept but an offset equals to log(0.05/0.95). The model is less sensitive than conventional tests for small D but similarly sensitive for large D. Visual tests are least sensitive overall. The model's curve lies between those of conventional and visual tests.~~

~~Finally, the power curves in Figure 8 are fitted~~ Figure 8 tells a similar story, with the fitted power curves illustrating how the sensitivity of each procedure changes with the magnitude of the departure. Following Majumder et al. (2013), we fit each curve using logistic regression ~~without intercepts and with~~ with a single slope in D and an offset of $\log(0.05/0.95)$~~to ensure a~~, fixing the rejection probability at 5% ~~rejection rate~~ under H_0. ~~The model's rejection curve consistently lies between those of conventional and human visual tests, indicating potential to further align its decision boundary with human visual assessment~~At every D the conventional test's curve is highest, the visual test's lowest, and the computer vision method's lies between them. In the non-linearity panel the visual test's curve does not reach one within the observed range, while the other two do.

5.2.3.

~~where $\bar{d}_{\text{true}}$ is the mean distance between the true residual plot and all~~

~~In the experiment conducted by Li et al. (2024), participants could select multiple plots per lineup. To reflect detection confidence, a weighted detection rate was computed for each lineup: selecting zero plots yielded a weight of 0.05; otherwise, if the true residual plot was selected, its weight was the inverse of the number of selections.~~



~~To evaluate the proposed distance measure, we could examine the relationship between the weighted detection rate and the δ-difference statistic from Chowdhury et al. (2018):~~

$$\delta = \bar{d}_{\text{true}} - \max_j \left(\bar{d}^{(j)}_{\text{null}} \right) \quad \text{for } j = 1, \ldots, m-1,$$

Figure 8: Rejection rates of visual tests, conventional tests and the computer vision method. Marks along the top of the plot represent rejections made by each procedure, and marks along the bottom represent acceptances. Curves are fitted by logistic regression with a single slope in D and no intercept but an offset equal to $\log(0.05/0.95)$. The method is less sensitive than conventional tests for small D but similarly sensitive for large D, and visual tests are least sensitive overall.

5.2.3. Adjusted δ-difference

The comparisons so far treat the computer vision method and the viewers as making a binary decision. Agreement between those decisions is one question, and whether the predicted distance measures what drives a viewer to select one plot over the others in a lineup is another. A score that ordered lineups by statistical significance alone could still agree with viewers while carrying no perceptual information of its own. We therefore ask whether the lineups that viewers detect more readily are the ones whose observed plot $\hat{D}$ separates most clearly from its null plots, ~~and $\bar{d}^{(j)}_{\text{null}}$ is the mean distance from null plot j to other nulls. These averages help address asymmetries in pairwise distances.~~ which is the quantity the adjustedδ-difference measures.

~~However, our method only compares each plot to a theoretically "good" residual plot , making the original formulation inapplicable. Moreover, since $D_{\text{null}} = 0$ by definition and D can not be derived from an image, we instead focus on $\hat{D}$, the model-estimated distance. This leads to the adjusted~~ Within a lineup, a viewer selects the plot that appears most different from the others, and a selection counts as a detection when the plot chosen is the observed one. The detection rate of a lineup is the weighted proportion of its

viewers who do so.[1] We measure how clearly the observed plot stands out from its own nulls by adapting the δ-difference ~~:~~ of Chowdhury et al. (2018) to the predicted distance,

$$\delta_{\text{adj}} = \hat{D} - \max_j \left(\hat{D}^{(j)}_{\text{null}} \right) \quad \text{for } j = 1, ..., m~~-1~~,$$

where $\hat{D}^{(j)}_{\text{null}}$ is the ~~estimated distance for~~ predicted distance of the j-th null plot ~~,~~and m is the number of ~~plots in a lineup~~ null plots. A negative δ_{adj} marks a lineup in which at least one null plot carries a larger predicted distance than the observed plot.

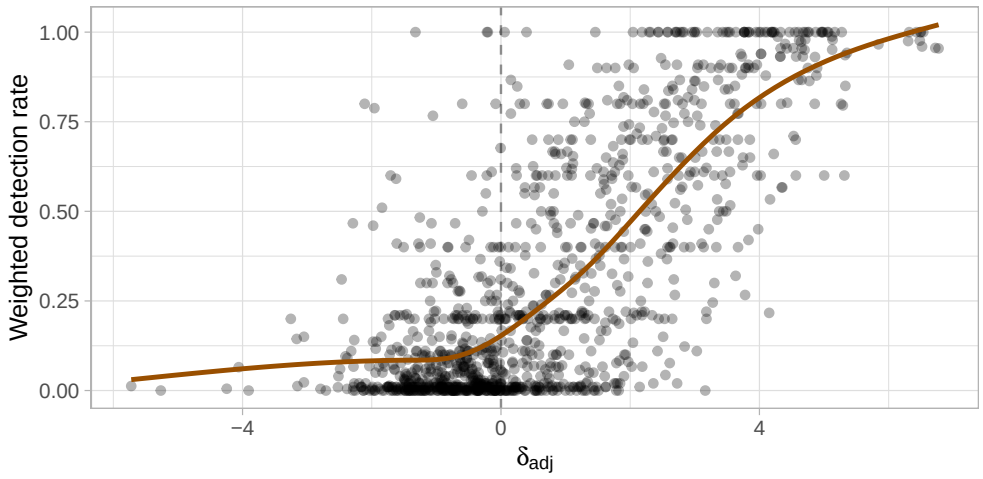


Figure 9: A weighted detection rate vs adjusted δ-difference plot. The brown ~~line~~ curve is a smoothing ~~curve produced~~ line fitted by ~~fitting~~ a generalized additive ~~models~~model. Detection ~~generally increases~~ rises with ~~positive~~ δ_{adj}, ~~but variability~~ and ~~exceptions highlight~~ the lineups at negative values are those in which at least one null plot carries a larger predicted distance ~~measure's imperfect alignment with human perception~~than the observed plot.

Figure 9 ~~displays the scatter plot of the~~ plots the weighted detection rate against δ_{adj}. ~~It shows that weighted detection generally increases with δ_{adj}~~ The rank correlation is 0.66 across the 1,~~especially when it is positive. This suggests that the differences in $\hat{D}$ capture some aspect of the visual separability between the true and null plots as perceived by humans, with larger positivedifferences generally indicating greater distinctiveness. However, the considerable variability in detection rates for similar~~ 116 lineups, and the mean detection rate increases from 0.09 in the lowest quartile of δ_{adj} ~~values implies that $\hat{D}$ does not fully account for all factors influencing human perception, and the alignment remains imperfect. A negative~~ to 0.13, 0.31 and 0.69 in the successive quartiles. The curve in Figure 9 is a generalized additive fit of the detection rate on δ_{adj}~~suggests some null plots may appear more visually distinctive to humans than the true residual plot. In some instances, the weighted detection rate is close to one despite a negative~~. It stays near 0.10 while δ_{adj} ~~. This discrepancy implies that the distance measure may not perfectly reflect actual human behavior~~is negative and rises steeply once it turns positive, so detection improves once the observed plot outranks every null plot. The predicted distance therefore broadly reflects how readily viewers distinguish the observed plot from its null plots.

The exceptions are consistent with that reading. Nearly half the lineups (45%) have a negative δ_{adj}, so at least one null plot ranks above the observed plot, but these have a lower mean distance ($D = 1.5$ against 3.8) and a lower detection rate (10% against 47%) than the rest, so a negative value mainly marks a violation too subtle to be seen rather than a failure to distinguish plots that are apparent.

[1] Participants could select more than one plot per lineup. A lineup in which no plot was selected receives weight 0.05, and otherwise, if the observed plot was selected, its weight is the inverse of the number of selections.

6. Examples

In this section, we present the performance of the ~~trained computer vision model~~ computer vision method on three example datasets, including the dataset associated with the residual plot displaying a "left-triangle" shape, the Boston housing dataset (Harrison Jr and Rubinfeld, 1978), and the "dino" datasets from the ~~from the~~ `datasauRus` package (Davies et al., 2022).

~~In the first case, both the model and human inspection correctly avoid rejecting H_0 when it is true, contrary to conventional tests, highlighting the necessity of visual examination. The second case shows a clear model violation detected by all tests, demonstrating the model's practical use for less intricate tasks. The third case involves an unusual deviation where the model and humans reject H_0, but some conventional tests do not, illustrating the benefits of visual-based decision-making.~~

6.1. Left-triangle

In Section 1, we introduced a residual plot (Figure 10A) where the "left-triangle" shape could be misinterpreted as evidence of heteroskedasticity. Although the residuals are drawn from a correctly specified model, the Breusch-Pagan test rejects H_0 ($p = 0.046$). This visual artifact stems from the skewed distribution of fitted values, as later illustrated in the lineup in Figure 13A, where similar "left-triangle" patterns appear across all residual plots. Since the ~~true~~ observed plot (at position 10) is visually indistinguishable from the null plots, a visual test would not reject H_0.

The computer vision ~~model~~ method yields the same conclusion from the lineup. As shown in Figure 10C, the ~~estimated distance $\hat{D}$ falls well below~~ predicted distance $\hat{D} = 1.06$ falls to about half the 95% quantile of the null distribution (2.18), with a p-value of 0.33. The bootstrapped distribution further suggests that rejection of the fitted model is highly unlikely. Together, the visual and ~~model-based~~ computer vision-based tests do not reject H_0, in contrast to the Breusch-Pagan test, which lacks the context provided by null plots and may be misled by visual artifacts.

To further interpret the computer vision model's output, the attention map in Figure 10B, computed as the gradient of the output with respect to the grayscale input, shows that the top-right and bottom-right corners of the residual plot have the most influence, corresponding to the triangle's vertices. In addition, a principal component analysis (PCA) of the 256 features extracted by the model's global pooling layer (Figure 10D) shows that null and bootstrapped plots cluster together in the space defined by the first two principal components, and the ~~true~~ observed residual plot is covered by the cluster. This overlap reinforces the conclusion that the ~~true~~ observed plot is visually indistinct, aligning with our interpretation of the lineup.

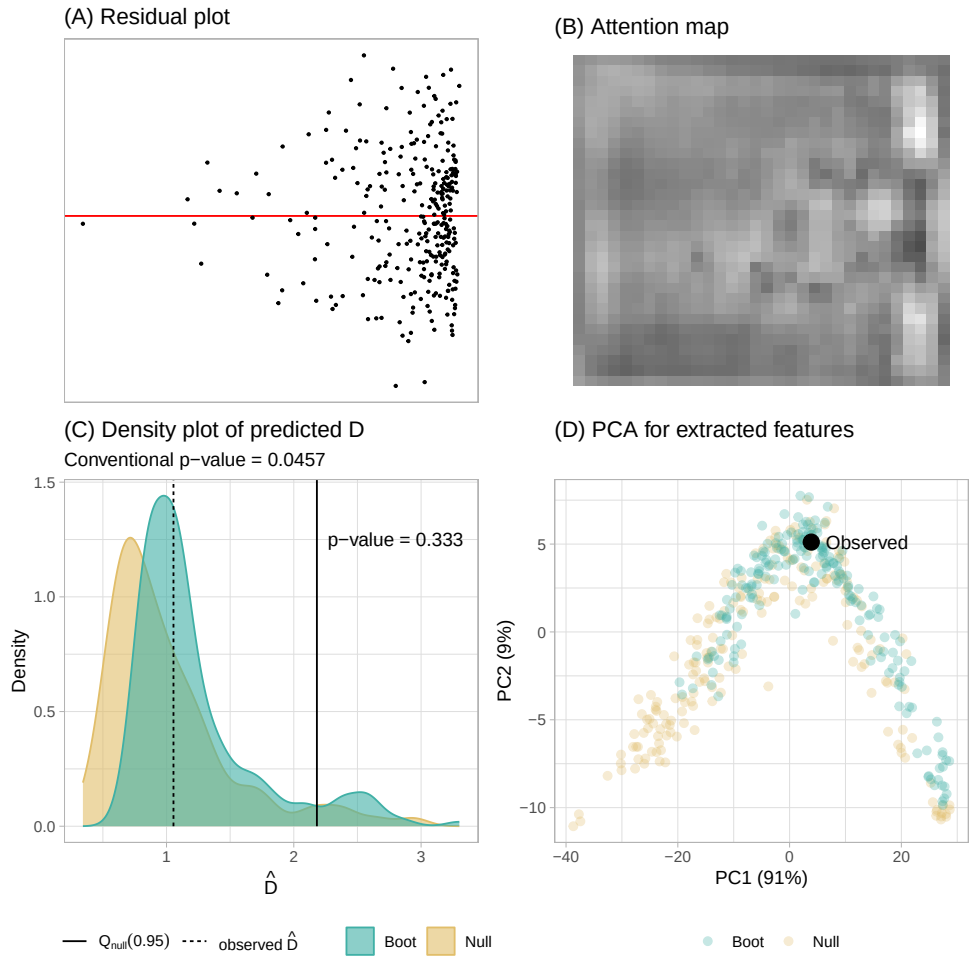


Figure 10: A summary of the residual plot assessment evaluated on 200 null plots and 200 bootstrapped plots. (A) The ~~true~~ observed residual plot exhibiting a "left-triangle" shape. (B) The attention map highlights the top-right and bottom-right corners of the residual plot as the most influential. (C) The density plot shows ~~estimated~~ predicted distances for null (yellow) and bootstrapped (green) plots. The fitted model is not rejected because $\hat{D} < Q_{null}(0.95)$. (D) Null and bootstrapped plots cluster together in the space defined by the first two principal components of the global pooling layer. The cluster also covers the ~~true~~ observed residual plot.

6.2. Boston Housing

The Boston housing dataset, originally published by Harrison Jr and Rubinfeld (1978), provides insights into housing in the Boston, Massachusetts area. For illustration, we use a reduced Kaggle version containing 489 observations and four variables: average number of rooms per dwelling (RM), percentage of lower status population (LSTAT), pupil-teacher ratio by town (PTRATIO), and median home value (MEDV). We fit a linear regression model with MEDV as the response and the remaining variables as predictors, focusing on detecting non-linearity, because the relationships between RM and MEDV or LSTAT and MEDV are non-linear.

Figure 11 summarizes the model evaluation. The residual plot (Figure 11A) reveals a distinct "U"-shaped pattern, suggesting non-linearity. The RESET test strongly rejects H_0 with a very small p-value, and the computer vision model yields a large ~~estimated distance $\hat{D}$, well above~~ predicted distance $\hat{D} = 6.48$, more than four times the 95% quantile of the null distribution (~~p-value = 0.00498~~)1.52), with p-value = 0.005. The bootstrapped distribution also shows widespread rejection, indicating the model is likely misspecified.

The attention map (Figure 11B) highlights the central region of the residual plot, corresponding to the turning point of the "U", as most influential. In the PCA projection (Figure 11D), bootstrapped and null plots form two distinct clusters, emphasizing the visual separability. This aligns with the lineup in Figure 13B, where the ~~true~~ observed plot stands out clearly. A human visual test would also likely reject H_0 in this case.

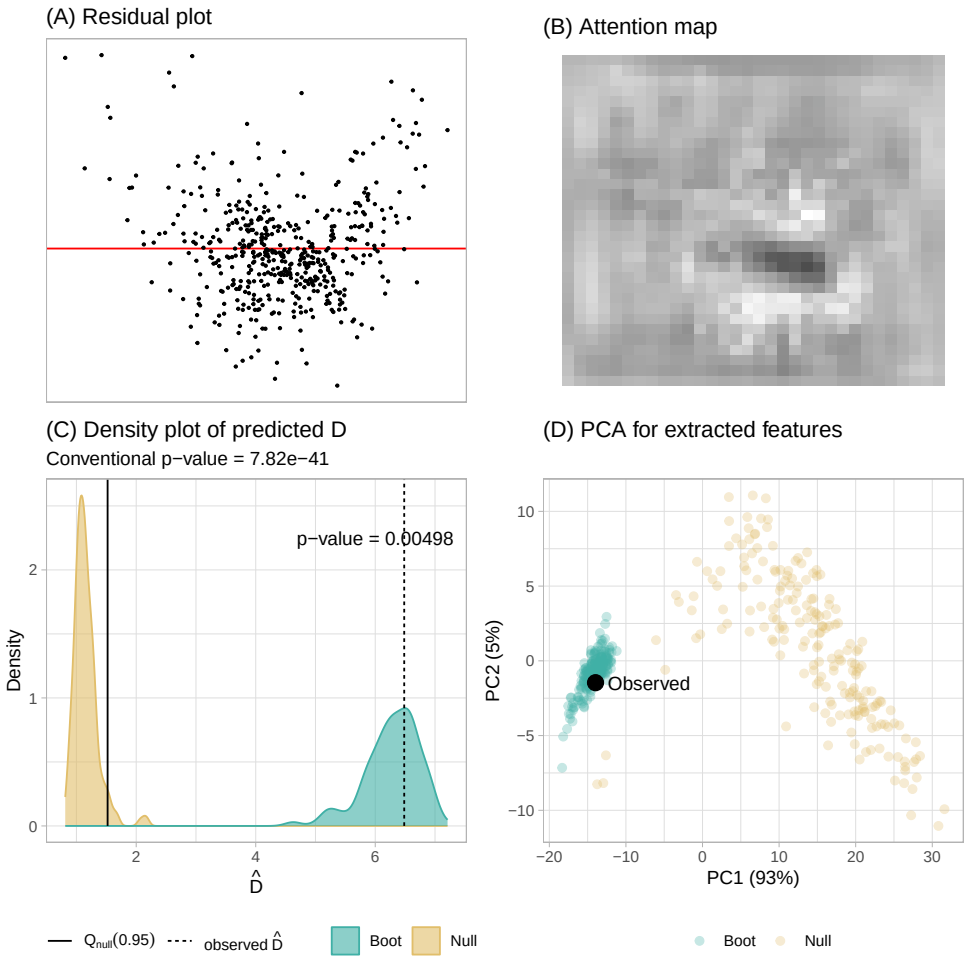


Figure 11: A summary of the residual plot assessment for the Boston housing fitted model evaluated on 200 null plots and 200 bootstrapped plots. (A) The ~~true~~ observed residual plot exhibiting a "U" shape. (B) The attention map highlights the central region of the "U" shape as the most influential part of the residual plot. (C) The density plot shows ~~estimated~~ predicted distances for null (yellow) and bootstrapped (green) plots. The fitted model is rejected because $\hat{D} \geq Q_{null}(0.95)$. (D) Bootstrapped and null plots form distinct clusters in the space of the first two principal components from the global pooling layer, highlighting their visual separability.

6.3. DatasauRus

Beyond ~~detecting common issues like~~ common departures such as non-linearity, heteroskedasticity, and non-normality, the computer vision model ~~can also identify unusual visual artifacts~~may also be sensitive to unusual visual patterns, such as shapes resembling real-world objects, ~~as long as such patterns do not appear in~~ particularly when these are absent from the null plots. These cases are challenging to categorize using conventional model diagnostics, so we compare the model's output with results from the RESET test, Breusch-Pagan test, and Shapiro-Wilk test (Shapiro and Wilk, 1965).

The "dino" dataset from the `datasauRus` R package illustrates this situation. It contains only two variables, x and y, but fitting a regression model produces a residual plot that clearly resembles a dinosaur (Figure 12A). This pattern is visually distinctive in the lineup shown in Figure 13C, where a human visual test would undoubtedly reject H_0.

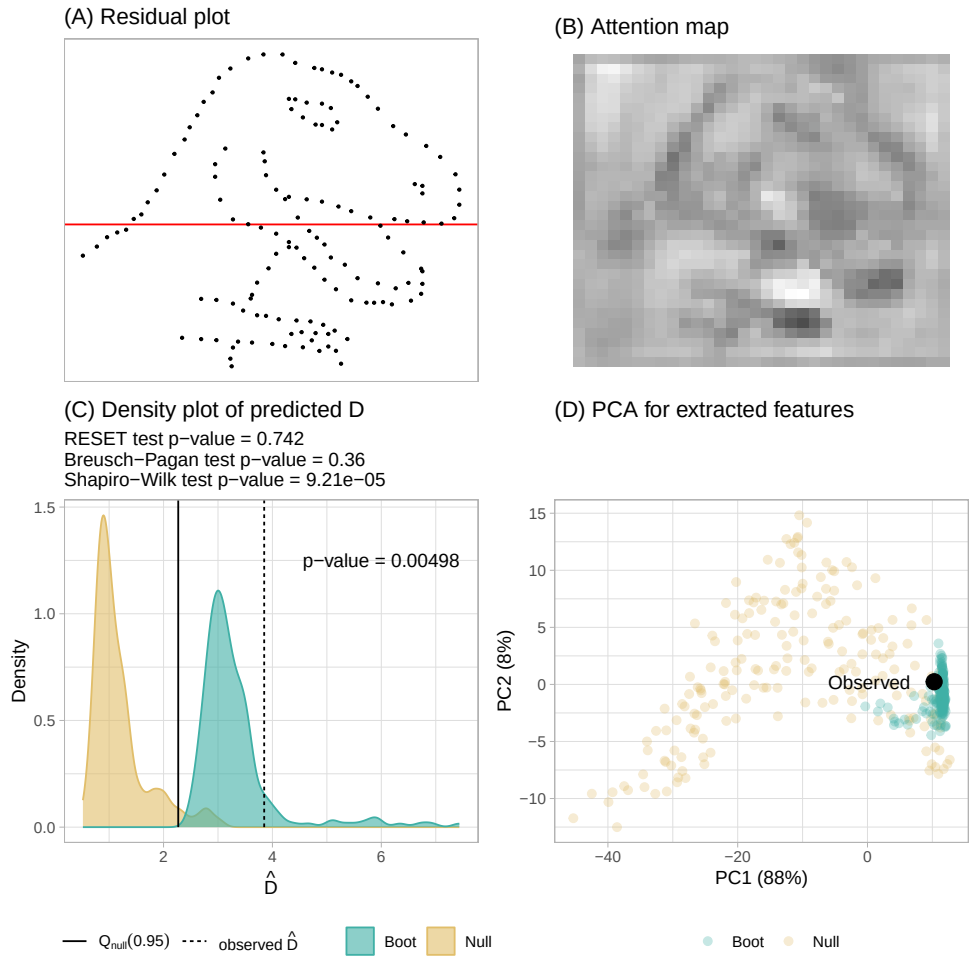


Figure 12: A summary of the residual plot assessment for the datasauRus fitted model evaluated on 200 null plots and 200 bootstrapped plots. (A) The residual plot exhibits a "dinosaur" shape. (B) The attention map highlights the "dinosaur" shape, indicating the model's decision is driven by human-recognizable features.(C) The density plot shows predicted distances for null (yellow) and bootstrapped (green) plots. The fitted model is rejected because $\hat{D} \geq Q_{null}(0.95)$. (D) The bootstrapped plots cluster at the corner of the null plot cluster, yet remain isolated in the space defined by the first two principal components of the global pooling layer.

Consistent with this, the computer vision model assigns a ~~distance $\hat{D}$~~ predicted distance $\hat{D} = 3.85$, above the 95% quantile of the null distribution (~~p-value = 0.00498),~~ 2.27), with p-value = 0.005, leading to rejection of H_0. The bootstrapped distribution further supports this, with most fitted models also being rejected. In contrast, the RESET and Breusch-Pagan tests yield p-values above 0.3, suggesting no violation. Only the Shapiro-Wilk test rejects normality with a small p-value.

Crucially, the attention map in Figure 12B highlights the dinosaur shape, indicating that the model's decision is driven by human-perceptible features. The model appears to capture contours and outlines in a manner similar to human visual interpretation. In the PCA plot (Figure 12D), the cluster of bootstrapped

plots is positioned at the corner of the cluster of null plots, isolated from the cluster of null plots.

In practice, such visual artifacts would be difficult to detect without inspecting the residual plot directly. Furthermore, analysts do not always test for normality, and even when violations are found, they are often deemed acceptable due to the robustness of linear regression under quasi-likelihood theory. This example highlights the value of visual diagnostics and the unique role of the proposed computer vision model in supporting them.

Taken together, the three cases illustrate how the computer vision method reproduces visual judgements without requiring a panel of human viewers. In the first case, a conventional test rejects the null but the method does not, agreeing with the visual assessment. Although the plot contains a triangle, the same feature also appears in the null plots generated for these residuals and fitted values, so the triangle does not provide sufficient evidence against the null despite being visually salient. In the third case, the method rejects the null even though two conventional tests fail to do so, again agreeing with the visual assessment. Here, the deviation is visually apparent, yet the test statistics fail to capture it. The second case provides a contrasting example in which all procedures agree.

These differences reflect what each procedure compares. A conventional test reduces the plot to a statistic designed to detect a particular type of departure and evaluates it against the corresponding null distribution. Its sensitivity is therefore limited to the departures that statistic captures. The computer vision method instead scores every plot in the same way and compares the score of the observed plot with those of null plots obtained by rotating the residuals. The rotations preserve the fitted values and residual magnitudes, so features arising from these quantities even when the fitted model is correct are also present in the null plots. Evidence against H_0 therefore comes from visual features that distinguish the observed plot from the null plots, much as a viewer judges the observed plot against the null plots in a lineup. The attention maps further suggest that the model focuses on visual features that a human viewer would also recognise, helping explain the agreement across the three cases.

~~A summary of the residual plot assessment for the datasauRus fitted model evaluated on 200 null plots and 200 bootstrapped plots. (A) The residual plot exhibits a "dinosaur" shape. (B) The attention map highlights the "dinosaur" shape, indicating the model's decision is driven by human-recognizable features.(C) The density plot shows estimated distances for null (yellow) and bootstrapped (green) plots. The fitted model is rejected because $\hat{D} \geq Q_{null}(0.95)$. (D) The bootstrapped plots cluster at the corner of the null plot cluster, yet remain isolated in the space defined by the first two principal components of the global pooling layer.~~

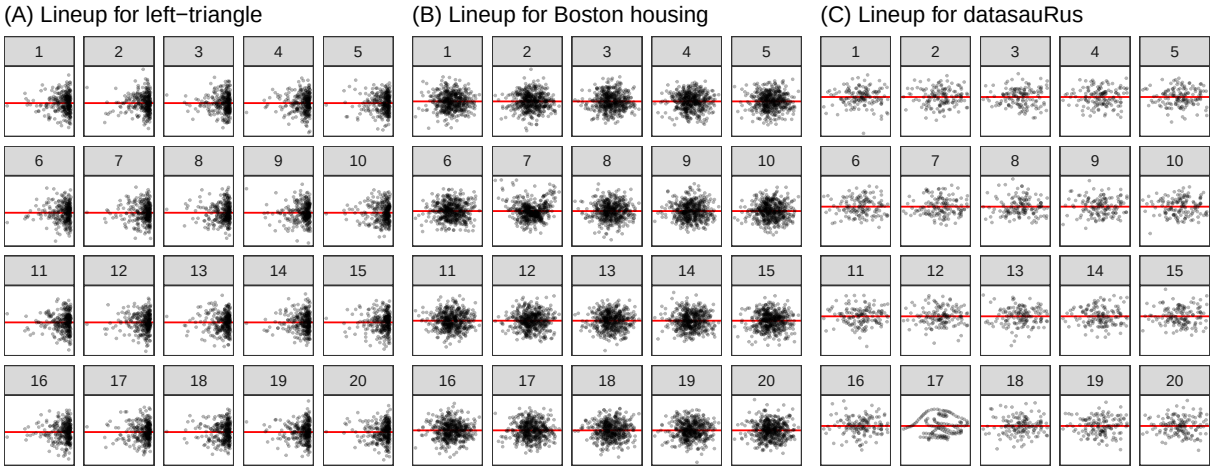


Figure 13: Lineups of residual plots for the "left-triangle", "Boston housing", and "datasauRus" datasets. (A) ~~True~~ Observed plot at position 10; no clear visual difference. (B) ~~True~~ Observed plot at position 7; "U"-shape clearly stands out. (C) ~~True~~ Observed plot at position 17; distinctly artificial and easily identifiable.

7. Limitations and ~~Future Work~~future works

~~While the~~ The computer vision model performs well under the synthetic data generation scheme~~and~~ , across the three illustrative examples, ~~this study~~ and on the lineup data. This study nevertheless has several limitations that point to directions for future work.

~~First, the~~ The proposed distance measure assumes that the true model ~~follows~~ is a classical normal linear regression ~~framework, which may be restrictive. Although we do not explore relaxation of this assumption here, future work could extend the approach to other types of regression models. One possibility is to define~~ model, which is restrictive. Extending it to other classes of regression model would take a distance measure tailored to each ~~model class and re-train the computer vision model accordingly. To reduce computational cost,~~ class and a model retrained on it, although the convolutional blocks ~~from our current model~~ could be reused ~~via~~ through transfer learning, ~~as it is already effective at capturing~~ since they already capture the relevant visual features. ~~Alternatively, transforming residuals to approximate~~ An alternative is to transform the residuals towards normality and constant variance~~could allow for meaningful comparisons under our current distance measure . However, if raw residuals are used, the interpretation remains valid only when the visual differences between the true and null plots are detectable by the proposed distance measure.~~ , potentially allowing the current measure to be applied to a broader range of models.

~~Second, this study focused solely on the standard residuals vs.~~ The distance is defined on $\mathcal{M}(P)$ and $\mathcal{M}(Q)$, so correlations between residuals are discarded and only their coordinatewise marginal distributions are compared. Dependence between residuals therefore goes undetected. Serial correlation, spatial correlation, repeated measurements on the same unit and batch or plate effects all produce residuals that are correlated while their marginal distributions appear ordinary. This limitation is less consequential for the residuals versus fitted values plot ~~,~~ considered here, which arranges residuals by fitted value rather than by the index or grouping carrying the dependence. This study also focuses on this single diagnostic plot, leaving out other common ~~diagnostic plots~~ displays such as residuals ~~vs.~~ versus predictors and Q-Q plots. ~~These alternative plots may offer additional information and could be incorporated into a broader visual testing framework in future research.~~ Addressing any of these would require both a distance defined on the joint distribution of the residuals and a display that carries the relevant structure, such as residuals against observation order, residuals by group, or a semivariogram.

~~Finally, we~~ The network is trained on the departures the simulation produces, over the ranges in Table 1, so outside that design nothing guarantees how closely its scores recover the ordering of D. That is a limit on what can be claimed rather than on what the model can do. Because the reference distribution Q defines what a good residual pattern looks like, the network can recognise departures beyond the simulated families. The dinosaur example in Section 6 illustrates this, since a pattern unlike anything in the training design is still scored as a clear violation. What is missing is a guarantee of performance in such cases.

The predicted distance is based on the rendered plot image, and the rendering configuration is fixed throughout this study. Consequently, the model's input depends on the chosen aesthetic mappings, layout, pixel resolution, and treatment of overplotting. Sample size must also be supplied separately, as it cannot be reliably recovered from the image alone. Changing the plotting style effectively changes the input representation, and the model's behaviour under a different style is unknown.

We chose a relatively simple computer vision architecture to establish a baseline for automated visual inference. While the current model shows promise, there is room to improve its alignment with human judgment. Achieving this may require more extensive external data from surveys or controlled experiments to better understand how human observers interpret residual plots and how these judgments differ from the model's outputs.

8. Conclusions

~~In this paper , we have introduced a distance measure based on the Kullback-Leibler divergence to quantify disparity between the residual distribution of a fitted~~ This paper turns a residual plot of a classical normal linear regression model ~~and the reference distribution~~ into a quantity that can be tested. The quantity is the distance of the residual distribution of the regression model from the residual distribution

expected under correct specification. ~~This measure captures the extent of model violations and forms the basis of a computer vision model that estimates the distance using residual plots as input~~A convolutional neural network then predicts it from the plot, and the predicted distance is compared with the predicted distances of rotations of the observed residuals, which yields a test with exact Type I error control. The network and the test together form the method that makes the assessment of a residual plot automatic, with the network supplying the reading a viewer would give and the rotations supplying the null distribution that reading is judged against.

~~The estimated distance supports a formal statistical testing procedure by comparing the true residual plot against null plots generated from the fitted model. Through bootstrapping and model refitting, we also evaluate how often a correctly specified model would be flagged as misspecified under repeated sampling~~We have proved that the validity of the rotation-based test does not depend on the network. Rotations leave the fitted values and residual magnitudes unchanged, making the observed plot and its rotated null plots exchangeable under H_0, regardless of the scoring function. Thus, the test maintains its nominal Type I error rate irrespective of the network's predictive accuracy. Predictive accuracy instead influences power. A poor predictor may fail to distinguish departures from the null, whereas a good predictor can detect them more effectively.

The ~~trained computer vision model performs well on both training and test data, though its performance is somewhat lower for non-linear patterns. Statistical tests based on the model's predicted distance show lower sensitivity~~ network predicts the distance accurately on plots outside its training set. Test error is only modestly higher than training error in every scenario, and predictions are most accurate for non-normality and least accurate for non-linearity. Applied to lineups from an earlier human-subject experiment, the method's rejection rate falls between those of conventional tests and the participants. For mild violations, the method rejects less often than conventional tests but ~~higher sensitivity than human visual tests. Importantly, the predicted distance broadly aligns with the strength of visual signals perceived by humans, though further improvements are possible~~more often than human viewers. As the departures become more pronounced, its rejection rate moves closer to that of conventional tests, while human viewers continue to reject less often. The predicted distance also broadly reflects how distinct the observed plot appears to viewers compared with its null plots. The margin between the predicted distance of the observed plot and the largest predicted distance among its nulls correlates moderately strongly with how often viewers select the observed plot from its lineup.

~~We demonstrate the method across several scenarios, highlighting its effectiveness and its conceptual alignment with visual inference. Both visual and distance-based tests act as omnibus procedures, detecting a wide range of model violations collectively. In contrast, conventional tests such as RESET and Breusch-Pagan target specific assumptions, making them less flexible and potentially harder to combine without adjusting for multiple testing.~~ The three illustrative examples demonstrate how the proposed method performs a form of visual assessment similar to that used by human viewers, without requiring a panel of viewers. By evaluating the visual features of the observed plot relative to those in the null plots, the method can discount visually salient features that also arise under the null, while identifying visual patterns that conventional tests are not designed to detect. The attention maps further suggest that some of the visual features contributing to the predicted distances are also relevant to human viewers.

~~Overall, our approach offers a practical tool to support regression diagnostics by reducing the manual effort required for residual plot inspection. While we recommend analysts to continue reading residual plots~~ The method is a screening tool rather than a substitute for visual inspection. It allows residual plots to be assessed at a scale that would be impractical for human viewers, without requiring human labels during training. The plot itself remains available to a reader, who can still recognise and interpret the patterns in it. We therefore recommend reading the plot whenever feasible, ~~our method can augment the process, either as an automated screening tool or a supplement to human judgment~~using the method's p-value to decide where closer inspection is most needed.

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The article was created with R packages `rticles` (Allaire et al., 2022), `knitr` (Xie, 2014) and `rmarkdown` (Xie et al., 2020). The project's GitHub repository (`https://github.com/TengMCing/auto_residual_reading_paper`) contains all materials required to reproduce this article.

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TODO

Data ~~Availability Statement~~availability statement

The data that support the findings of this study are openly available in the project's GitHub repository (`https://github.com/TengMCing/auto_residual_reading_paper`).

TODO: the training code

The human subject data used in this study were originally collected under ethical approval obtained by the original authors Li et al. (2024), who have made the data publicly available for re-use under an MIT license. As this study involves secondary analysis of de-identified, publicly available data, separate ethical approval is not required for the present analysis, in accordance with the journal's policy on secondary use of anonymized data.

Supplementary ~~Materials~~materials

Appendix: The appendix ~~includes more details about the synthetic data generation process, neural network layers used in the work, model training, and model violation index~~ records the hyperparameters that were searched when training the network and the protocol applied to each candidate, and it defines the model violations index together with the thresholds used to describe the strength of a violation. (appendix.pdf, Portable Document Format file)

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT (OpenAI, ~~GPT-4, accessed July~~ GPT-5.5, accessed Nov 2026) in order to improve the clarity and fluency of the manuscript text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. All content was reviewed and verified by the authors to ensure accuracy and integrity.

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